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MR603 -THESIS I

FROM PROTOTYPE TO PLATFORM: MEASURING THE COMMERCIAL VIABILITY OF DIGITAL ASSETS

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Abstract

Digital transformation has been so fast that software-based digital assets have become essential to organizations to innovate, become more efficient and stay competitive. A lot of software products might work in the prototype stage and a few of them may turn into a digital platform that can be scaled and profitable. While there are some studies looking at specific topics, such as software architecture, technical debt, API maturity, organisational maturity, platform governance and digital transformation, these topics have been studied independently of each other. At present, there is no complete framework that addresses the digital assets from technical, organisational and commercial perspectives, to evaluate their commercial viability.

The objective of this research is to develop and validate a framework, called the Commercial Viability Index (CVI), to assess software-based digital assets' readiness for transition from the prototype to scale and to a commercially viable digital platform. The study design is based on mixed methods of research which includes comprehensive literature review, development of conceptual framework, validation by experts and quantitative analysis. From managing technical debt to API maturity, platform governance and digital transformation capability, to commercial performance, and organisational readiness, the proposed framework covers all of the most important aspects that might influence whether a software architecture is commercially viable or not. The factors are evaluated and considered in terms of their impact on successful commercialisation of digital assets.

The anticipated outcomes from this research are a Digital Asset readiness validated Commercial Viability Index that will offer structured and evidence-based approach to assessing digital asset readiness. The framework is designed to assist organisations, investors, start-up founders, software developers and technology managers in determining their strengths and weaknesses and potential commercialisation risks before investing in the development of the platform and rollout to the market.

Regarding software engineering, information systems and digital innovation, the study is of practical significance, as it is an integrated assessment model that covers the technical and commercial aspects of success. The research also includes practical recommendations of what organizations can do to improve software commercialization and reduce investment risk, and

to encourage sustainable software commercialisation for digital platforms. The framework is likely to be a useful decision support tool for assessing and enhancing the commercial viability of digital assets in various sectors, including the booming digital economy in Australia.

Commercial viability measures, and the Commercial Viability Index (CVI), are important keywords in the scope of Digital Assets.

Platforms, Software Architecture, Platform governance, Technical debt, API Maturity, Organisational readiness, Digital transformation.

1.1 Chapter Introduction

Organisations have been dramatically impacted by the rise of digital technologies and how they have enabled them to generate value, provide services and compete in more digital economies. Once standalone pieces of software, the applications are increasingly becoming digital platforms that can accommodate multiple stakeholders, engage with the ecosystem and create sustainable commercial value. From a working software prototype to a scalable digital platform is a key goal for technology-intensive companies aiming for long-term competitiveness and innovation. The growing number of businesses that are pursuing digital transformation projects has resulted in research questions around what makes a digital asset successful in becoming a viable platform for trading [1].

Digital assets are software-based, cloud-based, mobile, application programming interfaces (APIs), artificial intelligence (AI) models, digital platforms and software-as-a-service (SaaS) that provide value to the organisation or business. Although numerous digital assets produce good technical results during the prototype phase, it is a relatively small number that make it to commercially viable platforms. Many software products have shown great potential and are struggling to be commercialized due to software architecture, technical debt, scalability, interoperability, organizational preparedness, governance, security, market uptake, and business strategy issues. The hurdles suggest that while technical prowess is not enough, it is crucial to have a good grasp of the market [2].

These are some of the individual technical aspects of software, that have been extensively studied in the software engineering literature: software quality, architectural design, maintainability, API maturity, technical debt management, software scalability. Likewise, research on information systems and digital transformation has focused on the issue of organisational capabilities, digital maturity, innovation management and platform ecosystem development. These research areas have, however, grown somewhat disconnected and there is a lack of integrated knowledge on the joint effect of technical, organisational and commercial challenges on the successful transition of digital assets to sustainable digital platforms.

To make the crucial decision of whether to invest heavy financial and organisational effort into developing a platform, organisations have had to consider the commercial readiness of

their software assets, which has been exacerbated by the growing significance of digital platforms. Evaluation methods today mostly rely on one or a few technical or financial metrics and are not comprehensive enough to consider software engineering quality attributes, organisational readiness/preparedness and commercial performance. As a result, organisations frequently find they don't have any systematic methods for figuring out if a digital asset has the characteristics that will enable its long-term commercial success.

This study aims to solve this issue by examining the factors that affect the commercialization of digital assets and developing a thorough framework to assess the maturity of a digital asset for adoption into a viable digital platform. The goal of the study is to combine software engineering principles, platform ecosystem theory, digital transformation, organisational readiness, API maturity, software technical debt management and commercial evaluation in a single framework that can be used by the research and industry community [3].

This chapter introduces background and context of the research, presents research problem and preliminary research gap, defines the research aim and objectives, formulates the research question, explains the significance and expected contribution of the study, outlines the scope and proposed methodology, discusses ethical issues, defines key terms, and finally the overall organisation of the thesis.

1.2 Background and Context

The speed of information and communication technology (ICT) development has changed the value creation, delivery and capture in an organization. In the last 20 years, digital technologies like cloud computing, artificial intelligence (AI), big data analytics, the Internet of Things (IoT), blockchain and application programming interfaces (APIs) have transformed software development processes and business models. Organisations are increasingly building digital platforms to allow continuous innovation, ecosystems to collaborate and scalable service delivery, as opposed to developing standalone software products for specific operational tasks. This change has transformed software into one of the most valuable

strategic assets in today's organisations and digital platforms into a key to economic growth and digital transformation [4].

Digital transformation is the use of digital technologies to improve the performance of organisations, their products, services, and/or business models. Digital transformation is not simply the adoption of new technologies, but rather a deep change in the organization that has an impact on the technology, business strategies, organizational culture and value creation mechanisms. Organisations are increasingly adopting a digital transformation, and software has gone beyond being a mere technical tool to becoming a strategic digital asset that can be responsible for creating value for the long term through innovation and platform-based business models.

A digital asset may be anything that is software-based and has measurable value generated by digital technologies. These encompass enterprise software systems, Software-as-a-Service (SaaS) Applications, cloud-native services, mobile applications, APIs, AI-powered solutions, and integrated digital platforms. Digital assets typically start off as a prototype that is created to address a particular technical or business issue. The main development tasks in the prototype phase are to verify functional requirements, demonstrate technical feasibility and receive initial user feedback. While numerous prototypes are able to accomplish these goals, relatively few moves on to become large-scale, commercially viable digital platforms that are able to sustain large populations of users and deliver sustained business value.

One of the most difficult steps in developing software is getting from a prototype to a platform. A software prototype is likely to present proof-of-concept functionality in a controlled environment while a digital platform will need to be scalable, interoperable, secure, reliable, maintainable, governed, and constantly evolving in highly dynamic business environments. It takes much more than good software engineering to make this transformation to success. In addition, organisations need to consider their commercial strategy, organisational readiness, developing the ecosystem, platform governance, customer adoption and operational sustainability in the long term.

Unlike traditional software products, modern digital platforms have a number of characteristics that make them unique. Modern digital platforms are very different from

traditional software products, and they enable interactions between a variety of stakeholders, such as developers, service providers, customers, business partners and third-party innovators. Platform-based business models generate value through the network-effect that the more that join a platform the more value it generates. It has been demonstrated by

companies like Microsoft, Amazon, Google, Salesforce, and Shopify that software platforms can be sustainable competitive advantages by expanding the ecosystem, integrating APIs, providing cloud services and continuously developing new features. These are just a few examples of how software capabilities are only part of the equation for commercial success; an organisation's experience of effectively managing platform evolution is just as crucial.

The software architecture is a basic element of whether or not a digital asset can become a business platform. The term software architecture describes the structure of a software system, its interactions between its components, communication patterns and methods, deployment patterns and strategies and scalability. Architectures that are well designed enhance software maintainability, flexibility, reliability It enhances and maintains performance, yet also cuts down long-term development costs. However, poor architectures can result in technical complexity, ongoing maintenance burden and restrict future scalability. When digital platforms grow, architectural decisions taken early in the software development process often play a key role in the product's commercial viability in the long term.

Technical Debt is another factor that affects the evolution of the platform. Technical debt is the debt incurred by making short term decisions in software development that result in software with poor quality, maintainability, and extensibility. In the prototype development phase, organizations often sacrifice masterpiece design for speedy delivery to meet market needs and to get the product to market as soon as possible. While this could spur initial innovation, as more technical debt is built up over time, it can lead to major complications when the platform scales up in the future due to the high cost of maintenance, lowered software quality, and slower development of new features. Good technical debt management has thus become an essential part of sustainable software development and digital platform evolution.

The use of Application Programming Interfaces (APIs) has also emerged as digital platforms' key enablers. APIs enable different software systems to interoperate with each other and allow other developers, partners and customers to use the platform services in a convenient manner. APIs are a key component in mature ecosystems that foster innovation, promote

platform adoption, and grow digital ecosystems via third-party application development. Many organizations with proper API strategies have been able to increase the scalability of platforms and the growth of ecosystems in the past as opposed to those based on proprietary software architectures.

Architectural quality, software maintainability, software technical debt, software scalability, software API maturity and software quality assurance are all topics that have been extensively studied in software engineering research, while digital transformation, organizational capabilities, platform ecosystems, technology adoption, and business innovation are the main topics studied in information systems research. These research streams have yielded useful knowledge on their own, but there has been surprisingly little research that connects these technical and organisational viewpoints into an all-encompassing framework, which can assess the commercial maturity of digital assets. Organisations then tend to use disjointed approaches to assess, examining individual technical, or financial, measures rather than taking a holistic view of the range of technical, organisational and commercial aspects that affect the success of the platform.

The subject is further highlighted by the fact that more and more investments are being made in digital innovation. Despite the success of a lot of software products in the technical sense, many of them are not able to reach commercial sustainability due to the investment that the organizations make in their development, which reaches billions of dollars annually. Various industry reports have confirmed that software failure is often caused by lack of scalability, poor software architecture, lack of software governance, lack of organisational readiness, poor commercial planning, lack of market alignment or failure to adapt to changing technological environments. This failure is costly, leads to loss of competitiveness for the organisation, postpones digital transformation projects, and inefficient use of development resources.

This research takes place in the context of the Australian digital economy. Many digital technologies are being rapidly embraced in government, healthcare, education, finance, manufacturing, and new technology sectors in Australia. Organisations are urged to

modernise legacy systems, embrace cloud computing, integrate AI technologies and create new and innovative digital services through national initiatives which drive digital transformation. But resource constraints, technical complexity, and organisational capability constraints are still challenging for Australian organisations, especially small and medium sized enterprises (SMEs), in commercialising software innovations. The creation of a systematic approach to evaluate the commercial potential of digital assets could thus offer beneficial guidance to technology developers, investors, policy makers and organisations looking to gain optimal return on their investment in digital innovation.

While the fields of software engineering, digital transformation, platform ecosystems and digital innovation have seen major advances in research, the intersection of all these technical and organisational issues is not yet known, and these are key factors in whether a software prototype is commercially viable to develop into a sustainable software platform. Current assessment approaches focus on different aspects of software quality or business performance and only offer an overall assessment of the commercial value. This restriction leaves organisations with uncertainties when making software investment decisions, assessing platform maturity, and allocating software development resources [5].

Based on this, this research focuses on the multidimensional factors that impact the commercial viability of digital assets, and proposes a holistic Commercial Viability Index (CVI) comprised of software architecture quality, API maturity, technical debt, organisational readiness, platform governance, digital transformation capability, and commercial performance indicators. The study introduces a framework for simultaneously evaluating the technical, organisational and business aspects of digital assets, providing a more structured solution to measure the readiness of software-based digital assets to successfully move from initial prototype to viable and sustainable digital platforms.

1.3 Problem Statement

Organisations have increasingly embraced digital technologies and invested heavily in digital assets, such as software and cloud applications, mobile apps, enterprise software, Artificial Intelligence (AI) systems and Software-as-a-Service (SaaS) solutions. A significant number of

these assets perform well in their prototype phase but very few make it through to commercialisation to become viable and scalable digital platforms. This means that it is not enough to be technically successful for the product to be commercially viable in the long-term

[6]. Technical aspects of software development (such as software quality, performance, security, maintainability and functionality) are the primary concerns in the existing software development practices. But other aspects—such as software architecture, scalability, API maturity, managing technical debt, readiness of organisations, platform governance, digital transformation capability and business strategy—must also be considered in successful commercialisation. While these have been taken up individually in previous studies, few take them into a comprehensive context of commercial readiness assessment [7].

Current methods of evaluation also have serious drawbacks. The major difference is that software engineering frameworks are concerned with the technical aspects, whereas business evaluation models are more concerned with the commercial aspects and potential of the market. Consequently, organisations have no structured process to decide if a software prototype is ready to be commercialised and evolve over time on the platform. This can result in wrong investments, high development expenses, small scale possibilities and less successful business outcomes.

Literature is still confined to software engineering, information systems and business management from an academic point of view. While the majority of the studies explores technical, organisational or commercial aspects separately, very few studies combine these aspects and develop a comprehensive digital asset commercial viability model. Moreover, few studies have been conducted on this aspect of successful commercialisation of platforms in general, and even fewer in the Australian digital economy.

It is, therefore, imperative to have a holistic framework that brings together software engineering quality, organisational readiness, platform governance, API maturity, digital transformation capability, commercial performance and technical debt together within a single framework. To fill this, this study suggests a tool called Commercial Viability Index (CVI)

which aims to systematically assess the readiness of software based digital assets to move from prototype to scalable and commercially viable digital platforms.

1.4 Preliminary Research Gap

With the development of digital transformation, a large amount of research has been conducted in software engineering, digital innovation, platform ecosystems and digital business models. Current research has focused on some key aspects as software architecture, software technical debt, API maturity, digital transformation, organizational readiness, platform governance, and commercial innovation. These studies have been useful to the understanding of software development and evolution of platforms but have focused mainly on the study of these phenomena in isolation and not as part of an integrated commercialisation process [8].

While software engineering research tends to be more technical, such as software quality, scalability, maintainability, and technical debt, information systems research tends to focus on digital transformation, capability of an organisation and platform ecosystems. Likewise, studies of business and innovation involve the analysis of market uptake, business models and business performance. While each research area offers valuable insights, there is a lack of research that integrates the technical, organizational and commercial perspectives in a holistic assessment of the commercial viability of digital assets.

Also, there isn't a common model to assess the commercial readiness of a project. Established frameworks tend to measure the quality of software or business performance, but not both. As a result, organisations frequently do not have a systematic approach to assessing if a software prototype is ready for becoming a scalable and commercially viable digital platform. In addition, there is a lack of empirical evidence of integrated commercialisation frameworks, especially in the context of the digital economy in Australia.

As a consequence of these limitations, there is a clear research gap. No holistic model exists that integrates software architecture quality, technical debt management, API maturity, organisational readiness, platform governance, digital transformation capability and

commercial performance in a systematic way to assess the commercial viability of digital assets.

In order to fill this lacuna, the present research suggests there is a need to develop a Commercial Viability Index (CVI) which incorporates technical, organisational and commercial aspects into a single assessment framework. The proposed framework is designed to offer a framework for organisations and researchers to measure and analyse the readiness of digital assets to move from prototype to a commercially viable and scalable digital platform, in a structured and evidence-based manner.

1.5 Research Aim

This research seeks to create and validate an in-depth framework called Commercial Viability Index (CVI) for assessing the readiness of software-based digital assets for transitioning from the prototype stage to a scalable and sustainable digital platform for commercialization. The proposed framework incorporates key technical, organisational and commercial aspects: software architecture quality, management of technical debt, API maturity, platform governance, organisational readiness, capability in digital transformation and commercial performance indicator, to offer a systematic and evidence-based method to assess the commercialisation of digital assets [9].

This research aims to develop a comprehensive evaluation framework that can be used for strategic decision making throughout the software development lifecycle (SDLC), as opposed to other evaluation approaches that are mainly based on individual technical metrics or business performance indicators. The study seeks to identify the critical factors which can affect successful platform evolution, explore the relationships between these factors and create a practical evaluation model, which can help software developers, technology managers, founders of start-ups, investors, and organisational decision makers to decide whether a digital asset is commercially ready for scaling up the platform. Finally, the research aims to provide theoretical knowledge and practical suggestions for enhancing effective commercialisation of digital assets in today's digital environments [10].

1.6 Research Objectives

The study will have the following aims to achieve the set research objectives, namely:

RO1: To critically analyse the current work on digital asset commercialisation, software architecture, platform ecosystems, technical debt, API maturity, organisations and digital transformation, and platform governance.

RO2: Identify and assess technical, organisational, and commercial considerations that affect the successful development of software prototypes to commercially viable digital platforms.

RO3: To understand the interplay between software architecture quality, technical debt, API maturity, organi Zational readiness, platform governance and commercial performance in supporting the commercial viability of digital assets.

RO4: To create a detailed Commercial Viability Index (CVI) matrix which incorporates the key parameters which affect digital asset commercialisation.

RO5: To test the proposed Commercial Viability Index framework through an appropriate research method (expert evaluation and empirical data analysis) to judge its reliability, applicability and effectiveness.

RO6: To make practical recommendations that will help software firms, tech start-ups, investors and decision makers assess and make progress in advancing the commercial viability of digital assets prior to extensive deployment on platforms [11].

1.7 Research Questions

The research questions have been formulated directly based on the research problem, preliminary research gap, research aim and research objectives presented in the previous sections. They are designed to inform the investigation by looking at technical, organisational and commercial considerations affecting the successful commercialisation of software-based digital assets. The questions also serve as the basis for choosing an appropriate research

methodology and assessing the proposed Commercial Viability Index (CVI) framework. Main Research Question

MRQ:

What are the key factors that enable the systematic measurement of the commercial viability of software-based digital assets to assess their maturity for the next step towards becoming scalable and commercially viable digital platforms?

This is the central purpose of the study and the main research question, which concerns the development of a comprehensive evaluation framework, considering technical, organisational and commercial aspects of digital asset commercialisation.

Supporting Research Questions:

The following research questions are put forward to answer the main research question:

RQ1: What are the most significant technical, organisational and commercial issues that are important for the successful commercialisation of digital assets?

The purpose of this question is to determine the major factors that are important to the successful development of software prototypes to commercially viable digital platforms. They are related to the software architecture quality, the management and control of technical debt, API maturity, preparation of the organisation, platform governance, digital transformation capability, and commercial performance.

RQ2: What is the interaction between software architecture quality, software technical debt, API maturity, organisational readiness, and platform governance in terms of commercial viability of digital assets?

This question explores the inter-relationships between the main technical and organisational variables found in the literature. These factors are not considered individually, but instead are explored in the context of their interaction to determine how these might affect scalability of the platform and its commercial sustainability over time.

RQ3: What is the best way to incorporate these multidimensional factors into a new framework of Commercial Viability Index (CVI) to evaluate digital asset readiness?

The question is tied to the design and development of a practical assessment framework, which can be used for a systematic evaluation of the commercial readiness of digital assets that are software based and that can be evaluated based on measurable evaluation criteria.

RQ 4: What is the effectiveness of the proposed Commercial Viability Index (CVI) framework in assessing the commercial viability of digital assets, relative to alternative approaches to assessing commercial viability?

The reliability, applicability and practicality of the proposed framework is empirically validated and expertly assessed in the last research question. The results will help to assess if the existing framework offers a more complete and effective way to evaluate commercialisation of digital assets than the currently available stand-alone assessment methods.

The research questions together set a clear direction for the research and guarantee coherence between the research gap identified, the research objectives and the proposed research methodology. These questions and their responses will help to guide the creation of a comprehensive and formal framework for evidence-based decision-making in developing digital platforms by software developers, technology firms, investors, and researchers [12].

1.8 Research Hypotheses

In this study, a framework development and validation (FD and V) approach using a quantitative research method is adopted and hypotheses are formulated to explore the relationships among major constructs which affect the commercial viability of digital assets. These hypotheses will be checked for their validity in an empirical manner in the data analysis stage of the research.

H1

The quality of software architecture significantly and positively affects the commercial viability of Digital assets.

Good software architecture enhances scalability, maintainability, interoperability and long-term evolution of the system, and consequently increases the chances for software prototypes to be successfully transitioned to commercial digital platforms.

H2

Technical debt is a negative and significant impact on digital assets commerce.

A high degree of technical debt leads to higher software maintenance fees, lower quality software, and lack of scalability, and less commercial readiness.

H3

The maturity of APIs has a positive and significant impact on commercialisation of digital assets.

APIs that are well designed and have become fully developed promote interoperability, ecosystem involvement, third party integration and platform scalability and therefore add to commercial success.

H4

The business potential of digital assets is associated with a positive organisational readiness.

Organisations with the right digital capabilities, leadership support, and trained workforce along with an aligned strategy are more likely to be successful in commercialising software products. H5

A good platform governance has a positive impact on digital assets' commercial viability.

Proper governance models promote coordination of ecosystems, collaboration amongst stakeholders, security, compliance, and sustainable ecosystems over time on the platform.

H6

The ability to transform digitally has a positive effect on commercialisation of digital assets.

Organisations that have a higher level of digital transformation maturity will be more likely to be able to leverage software innovation to create scalable digital platforms that create sustainable commercial value.

H7

The proposed Commercial Viability Index (CVI) will be considered more decision-useful than a predefined simpler comparator only if measurement-model validity, collinearity diagnostics, baseline comparison, and practical or expert validation provide adequate supporting evidence. H7 is therefore treated as a validation proposition rather than as an already supported claim.

1.9 Research Significance

The importance of this research lies in the fact that it fills the gap in the previous studies and provides a holistic approach with respect to the commercial feasibility of software based digital asset. The current evaluation methods are mostly devoted to the technical performance or business outcomes, and seldom combine technical performance and business outcome into a single evaluation model. This is where the proposed Commercial Viability Index (CVI) comes in to play, as it will help to integrate technical, organisational and commercial considerations, thereby enabling better decision-making when it comes to digital asset commercialisation.

1.9.1 Academic Significance

This study brings together software architecture, software technical debt, API maturity, organisation readiness, platform governance and digital transformation within one evaluation framework, enriching the area of software engineering, information systems and digital innovation with new approaches. This research offers a comprehensive analysis on the commercial readiness of digital assets as opposed to the previous ones which looked at the factors individually. The proposed framework is also a base for future research for digital platform evolution and software commercialisation.

1.9.2 Practical Significance

The proposed Commercial Viability Index (CVI) offers a operational decision support tool for software organisations, start-ups, investors as well as technology managers. It allows organisations to test the viability of software products before large-scale investments in software platforms. The framework can be used to identify technical and organisational weaknesses, minimize the risks associated with commercialization, optimize resource utilization and help in strategic planning. It can also help policy makers and innovation agencies to facilitate successful digital transformation projects, such as in the Australian digital economy.

1.9.3 Technical Significance

Technically, this research creates a comprehensive evaluation framework that will combine the software architecture quality, API maturity, technical debt management, organisational readiness, platform governance, and commercial performance, into a single assessment model. It is hoped that the framework will help facilitate evidence-based decision making, give software greater scalability and maintainability and help software development teams prepare their digital assets for successful commercialisation. Moreover, the methodology proposed can be transferable from software domain to software domain and thus can be used for different digital products and/or platform-based systems.

In terms of academics, the study is useful because it offers a systematic method to evaluate the commercial viability of digital assets and to facilitate their transition from software prototype to scalable and commercially viable digital platforms. In terms of practical, the research is valuable as it provides a system to evaluate commercial viability and to assist in making transition from a software prototype to a scalable and commercially sustainable digital platform. In terms of technical, the research is useful because it provides a system to evaluate commercial viability, to assist in the transition from a software prototype to a scalable and commercially sustainable digital platform.

1.10 Proposed Research Contribution

The research will be of great value to academia, industry and technology, as it will create a complete system to assess the commercial viability of software-based digital assets. This study differs from existing literature which tend to focus on technical, organisational or commercial issues independently by combining these elements in a single Commercial Viability Index (CVI) to evaluate the readiness of digital assets to move from prototypes to scalable and commercially viable digital platforms.

This work is important because it is the first research to define and develop a tool for evaluating commercial viability, the Commercial Viability Index (CVI), which integrates software architecture quality, technical debt management, API maturity, organisational readiness, platform governance, digital transformation capability and commercial performance into one single assessment framework. This integrated approach expected to deliver a more systematic and evidence-based evaluation of digital asset commercialisation than current stand-alone approaches.

Academically, it is an extension of the existing knowledge by integrating the concepts of software engineering, information systems, and digital innovation literature. It gives a multidisciplinary framework that puts together the interplay of the three aspects technical, organisational and commercial on platform evolution and commercial success.

In a practical sense, the proposed framework provides an organized decision-support mechanism to check if a technology is ready for commercial application prior to substantial investments in platform development for organizations, software developers, startup founders, investors and technology managers. This can help to minimize commercialisation risk, facilitate strategic planning and better resource allocation.

Last but not least, the research will yield an empirically validated model which can be transferred to other software contexts such as cloud computing, SAAS, AI software, enterprise software, fintech and health care software and other software for digital platforms. The results will enable researchers to use the methodology for future research and to enhance the commercialisation of software products.

1.11 Scope and Delimitations

This research is grounded on the analysis of commercial viability of software supported digital assets and creation of a framework to assess the readiness of commercial platform evolution of digital assets. This study looks at technical, organisational and commercial issues, that impact the successful implementation of software prototypes to scalable and sustainable digital platforms. The technical scope includes the software architecture quality, technical debt management, API maturity, scalability and platform governance. The organisational scope takes in digital transformation capability, organisational readiness and governance, and the commercial scope covers commercial performance and platform sustainability. The quantitative evidence in the current thesis stage is a secondary 250 observation benchmark that is part of the European Customer Satisfaction Index (ECSI) sample project recorded by SmartPLS and first described by Tenenhaus et al. (2005). It is not offered as a new and up-to-date survey of Australian practitioners. Therefore, any thesis specific CVI constructs or indicators must be mapped to the source variable in an explicit way, prior to making substantive CVI claims. The study is confined to digital assets (software) and excludes physical products and non-digital commercial assets. It does not try to make long-term financial projections. The conceptual motivation is as relevant to the Australia context of the digital economy as the secondary benchmark data that are currently available are from the context of mobile phones in France; therefore, population generalisation into Australia is not within the evidential scope of the present data. The study does not involve implementation of a commercial software platform, but rather, it aims at the development and validation of the framework.

1.12. Overview of the research methodology

This study takes a pragmatic (quantitative) position for the development of the framework. The proposed dimensions and hypotheses of CVI were defined by literature synthesis and the main empirical method for testing the measurement and structural models was PLS-SEM. Qualitative expert evaluation is planned and is not reported as a completed practical

validation, even if expert evidence exists, it is considered "not completed". This is the empirical benchmark at the present time: a sample project of 250 observations publically reported, ECSI. The original ECI model was designed for use in the study of customer satisfaction, so the variables in the original model cannot be automatically renamed as the seven CVI constructs. The benchmark can then only be measured as direct evidence for software architecture quality, technical debt, API maturity, organisational readiness, platform governance, digital transformation capability, and commercial performance if a reproducible mapping has been carried out or an independently documented thesis instrument. Seven key constructs from literature, such as software architecture quality, software technical debt management, API maturity, organisational readiness, platform governance, digital transformation capability, and commercial performance will be combined, and a Commercial Viability Index (CVI) will be created. These constructs will be utilized to build the conceptual model that will be tested through an empirical method to investigate their effect on digital assets' commercial viability. The descriptive screening and preliminary reliability checks can be performed using IBM SPSS, the PLS-SEM can be performed using SmartPLS 4. The measurement sequence of interpretation is started with indicator loading, then internal consistency reliability, convergent validity (AVE), discriminant validity (including HTMT), and finally, collinearity (VIF) are checked. For inferential assessment, the bootstrapping with 5,000 resamples is set. If the discriminant validity and/or collinearity is unacceptable, then the path coefficients and H1-H6 are considered provisional. Finalization of the CVI will not take place until the retained constructs show acceptable measurement properties. A transparent equal-weight or simple-score baseline will be calculated, and compared to any PLS-derived weighting approach for stability, interpretability and sensitivity. Prior to aggregation, it is also important to have a directionally documented Technical Debt score, where higher scores always indicate debt burden or debt-management capability.

1.13 Ethical Considerations

The 250 observations currently used as the ECSI benchmark are a pre-existing secondary data set and the thesis does not claim new human-research ethics approval or new participant recruitment, consent and response rate for the historical observations. The source and

context of the benchmark is clearly documented and academic use is clearly defined. New questionnaires, interviews or expert validation activities (which are specifically related to this thesis) should be approved by the relevant institutional ethics committee and informed consent obtained prior to data collection. If additional information is gathered from experts and/or participants in a subsequent validation phase, this will be conducted voluntarily, participants will be provided with relevant study information, informed consent and confidentiality will be observed according to institutional standards. Secondary data and any subsequent primary data will be responsibly stored and analysed. Personally identifiable information will not be reported and the thesis will clearly identify the original benchmark variables, transformations/mappings of these variables, and the analytical outputs produced for conducting the CVI. The research will be conducted in a way that is honest, transparent and will follow academic integrity. All sources of information will be cited correctly in the referencing format provided by the IEEE and the findings of the research carried out will be presented correctly without any fabrication, falsification or plagiarism. These ethical practices will make sure that the research is done in an ethical way, and that the results of the research are trustworthy and credible.

1.14 Definition of Key Terms

To ensure consistency throughout this research, the following key terms are defined as they are used in this study.

Digital Asset:

A digital asset refers to a software-based resource that creates organisational or commercial value through digital technologies. In this research, digital assets include software applications, cloud services, mobile applications, APIs, Software-as-a-Service (SaaS) solutions, artificial intelligence applications, and digital platforms.

Commercial Viability:

Commercial viability refers to the capability of a digital asset to achieve sustainable commercial success by satisfying technical, organisational, and market requirements while generating long-term value.

Commercial Viability Index (CVI):

The Commercial Viability Index (CVI) is the proposed evaluation framework developed in this research to measure the readiness of software-based digital assets for successful commercialisation. The framework integrates technical, organisational, and commercial factors into a single assessment model.

Digital Platform:

A digital platform is a software-based ecosystem that enables interactions among users, developers, organisations, and third-party service providers through shared digital infrastructure and services.

Technical Debt:

Technical debt refers to the long-term consequences of choosing short-term software development solutions that reduce software quality, increase maintenance costs, and limit future scalability.

API Maturity:

API maturity represents the level of quality, standardisation, security, documentation, and interoperability provided by Application Programming Interfaces that support software integration and ecosystem development.

Platform Governance:

Platform governance refers to the policies, standards, processes, and management mechanisms used to regulate interactions, security, participation, and decision-making within a digital platform ecosystem.

Organisational Readiness:

Organisational readiness describes an organisation's capability to successfully implement and commercialise digital innovations through appropriate leadership, digital skills, resources, organisational culture, and strategic alignment.

1.15 Organisation of the Thesis

This thesis is organised into five chapters to provide a logical progression from the research problem to the final conclusions.

Chapter 1 – Introduction presents the research background, problem statement, preliminary research gap, research aim, objectives, research questions, significance, proposed contribution, scope, methodology overview, ethical considerations, and the overall structure of the thesis.

Chapter 2 – Literature Review critically reviews the existing literature related to digital asset commercialisation, software architecture, technical debt, API maturity, digital transformation, organisational readiness, platform governance, and digital platforms. The chapter identifies limitations in current research and establishes the research gap addressed by this study.

Chapter 3 – Research Methodology explains the research philosophy, research design, conceptual framework, hypotheses, sampling strategy, data collection procedures, measurement instruments, and data analysis techniques used to develop and validate the proposed Commercial Viability Index (CVI).

Chapter 4 – Results and Discussion presents the findings obtained from the empirical analysis, evaluates the proposed hypotheses, validates the Commercial Viability Index, and discusses the results in relation to previous studies and the research questions.

Chapter 5 – Conclusion and Recommendations summarises the major findings, highlights the academic and practical contributions of the research, discusses the limitations of the study, and provides recommendations for future research and practical implementation.

1.16 Chapter Summary

This chapter introduced the research by presenting the background and context of digital asset commercialisation and explaining the importance of evaluating the readiness of software-based digital assets for commercial platform development. It identified the research problem and highlighted the lack of an integrated framework for assessing commercial

viability. A preliminary research gap was established, followed by the research aim, objectives, research questions, and hypotheses that guide the study.

The chapter also discussed the academic, practical, and technical significance of the research, outlined the expected research contributions, defined the scope of the study, and provided an overview of the proposed research methodology and ethical considerations. Finally, key terms were defined, and the overall organisation of the thesis was presented.

The next chapter provides a comprehensive review of the literature related to software architecture, digital platforms, technical debt, API maturity, organisational readiness, platform governance, and digital transformation. It critically analyses previous studies, identifies the major limitations in existing research, and establishes the research gap that forms the foundation for the proposed Commercial Viability Index (CVI) framework.

2.1 Chapter Introduction

Digital technologies are changing the way organisations design, develop, deploy and commercialise software products in a dramatic way. Today's software systems are no longer standalone, single-user applications but are more and more being engineered as digital platforms that enable customers, software developers, business partners and third-party service providers to interact with each other within a network of software systems. Due to organisations large investments in cloud computing, Artificial Intelligence (AI), Internet of Things (IoT), microservices, APIs, DevOps, and platform-based business models, the question of how software products are successful at maturing from technical prototypes to commercially viable digital platforms have become an exciting research topic [13] [14].

There have been many studies conducted on software architecture, digital transformation, organisational readiness, platform governance, API ecosystems and software quality, but the majority of these have focused on a single aspect of each of these. In the literature on software engineering, the emphasis has been on enhancing software quality in terms of its architectural design, maintainability, scalability, reliability, and code quality. Information systems and digital business, on the other hand, have focused on organisational capability, innovation management, digital transformation, governance mechanisms, and on business strategy. Thus, there is a lack of studies that have combined these multidisciplinary viewpoints in a coherent approach that could assess the commercial potential of digital assets as they go from prototype to platform [15].

It has also been shown in the literature that software projects often are technically successful but not commercially successful. A significant number of prototypes have good software quality and good functional performance but fail to mature to viable platforms, due to governance issues, organisational capacity, lack of ecosystem involvement, interoperability, API maturity, and maintainability. From these observations, it could be concluded that there are many more factors involved than just the technical quality of the software in the commercial success. Current research thus shows the need for a multi-dimensional evaluation framework that consider technical, organizational, governance and ecosystem and commercial factors.

The main concepts, which form the basis for the proposed research, are reviewed in this chapter. It starts by exploring the background and development of digital assets and digital platforms and the key concepts that are used throughout the thesis. The chapter then examines theory underpinning the research such as Platform Ecosystem Theory, the Resource Based View (RBV), and Dynamic Capabilities Theory. Further research is presented within a critical review of the literature of software architecture, technical debt, API maturity, platform governance, digital organisational readiness and commercial viability.

The review is not a discussion of these topics but rather a critical comparison of the previous studies in order to reveal their common findings, conflicting views, and unanswered questions. As existing literature indicates that technical and organisational characteristics of the software work together to affect commercial results, but they are not studied simultaneously, particular attention is given to the interaction between these two. The review also investigates the existing commercial tools and methods for evaluation and reveals strengths and weaknesses of these tools in evaluating software-intensive digital assets [16].

The last part of the chapter brings together the literature reviewed to see the main research gap that will be addressed in this thesis. This synthesis serves as the theoretical basis for the creation of the proposed Commercial Viability Index (CVI), an overall index for assessing the commercial readiness of digital assets, as they move from prototype to platform. The suggested framework integrates the technical, architectural, governance, organisational, ecosystem and commercial aspects, aiming to overcome existing evaluation framework limitations and provide theoretical and practical insights to the fields of software engineering and digital platform evaluation research [17].

2.2 Background and Context of the Research Area

In the last 20 years, the commercialisation of digital assets has been one of the most important research fields in the domains of software engineering, information systems and digital business. Organisations have been fundamentally changed in how they create, deliver and capture value through software with the advent of rapid advances in the fields of cloud computing, AI, big data analytics, Internet of Things (IoT), blockchain technologies, microservices, and Software-as-a-Service (SaaS) [18]. In contrast to the software products

that were traditionally created for internal use within an organisation or sold as a standalone product, today's software has become digital platforms designed for a variety of different stakeholders such as customers, developers, business partners and third-party service providers. This shift has turned software into a new tool for the organisation that can be used as a business asset to create a digital ecosystem with commercial value at all times [19].

The old way of software development was mostly project-based with software applications being developed to meet the needs of the organisation, and then released as a finished product, with little or no expectation for further evolution. Early software engineering methodologies were based on the Waterfall model, which focused on sequential software development, on the documentation of the software, and on the requirement up-front. Such techniques were effective at increasing software reliability, but were not very flexible when dealing with dynamically changing market requirements and customer needs. With technology and business environments becoming more dynamic, organisations needed to be more adaptive, with software development methodologies that were able to deliver continual innovation and quicker time-to-market.

Software Development cycles have been accelerated by introducing the methods of Agile, DevOps, Continuous Integration and Continuous Deployment. Organisations were able to now get functional software prototyping quickly, get feedback from the users, and improve the product by going through iterations. So, the idea of the software prototype was extended from ‘just a proof that it's technically possible’. In many cases, modern prototypes are the first part of a larger digital platform project, and can be used to test the concept of the business model, obtain investment, and create user communities before releasing the platform for commercial purposes.

However, while these technological developments provide software to be successful in creating a sustainable digital platform is much more complex than creating the software prototype. Many software start-ups manage to successfully introduce and demonstrate innovative technologies at the early development phase, but don't get to sustainable commercial scale once they've made it into the market. Likewise, numerous enterprise digital

transformation projects result in systems that are technically viable, but not expected to create business value because of lack of 'organisational readiness', governance ability, ecosystem development, interoperability and strategic alignment. The above observations show that while technical function is not enough for commercial success, it is certainly a prerequisite.

With the growth of platform business model, software commercialisation has been totally reinvented. The value of digital platforms is significantly different from the value added by conventional software products which are mostly used by customers, since they create value through interactions among different groups of participants. Examples include Cloud Service Providers, Online Marketplaces, Financial Technology Platforms, Healthcare Information Systems, Educational Technology Platforms and Enterprise Software Ecosystems. In those use cases, as well as in any other, the quality of the software is essential, but it isn't enough, since the software must also be capable of attracting developers, integrate third party services, meet governance standards, interoperable, and continue to evolve as per stakeholder needs.

As a result of this shift, there are increasing academic studies on digital platform ecosystems. Platform ecosystem research states that platform software ought to get the competitive benefit not only from its internal technical capacities, but additionally from network impacts, complimentary innovation and the creation of worth collaborations. The more members/stakeholders that join a platform ecosystem, the more value the platform would have for all the participants. As a result, the maturity of APIs, platform governance, interoperability, security, quality of documentation, and ecosystem management are now considered more and more as key factors for the commercial sustainability of an API.

Along with the platform ecosystems has come great changes in software engineering research. While correctness, reliability, maintainability, portability, efficiency are still relevant, scalability, architectural flexibility, modularity, resilience, interoperability, and evolution are becoming more of a focus in modern software engineering practices. The microservices era has ushered in a new approach to software design – one that moves away from traditional monolithic applications and towards loosely coupled services which can be

deployed, continuously delivered and maintained independently. While microservices enhance scalability and agility in an organisation, they also add to the operational complexity, governance needs of distributed systems and management challenges. As a result, aspects of architectural design now affect more than just technical performance, and are becoming more important for the long-term commercial viability of the building.

Technical debt is another key advancement, and a fundamental topic in software evolution studies. Technical debt refers to the expenses that will accrue to the software in the future, due to design compromises or implementation shortcuts that were used to get the software to market. Compromises might lower development expenses in the short run but technical debt can lead to a higher maintenance burden, lower flexibility in software, and limit further innovation. Often, companies that work in large competitive digital markets take a shortcut to getting their products to market quickly, and end up with technical debt that can restrict the growth of their platforms and drive up their costs. Technical debt is thus no longer just a software engineering issue, it's a strategic one now.

Cloud computing has opened the door to commercial software to all as well. Organisations can take advantage of cloud platforms to leverage scalable infrastructure, on-demand computing resources, and availability of services worldwide, to commercialise digital products more efficiently than the traditional infrastructure models. Cloud-based software systems are elastic, scalable and automated.

It is able to be deployed and quickly integrated with external services via APIs. Cloud computing has therefore contributed greatly in lowering the barriers to entry in the market as well as competition among software vendors. Organisations need to therefore stand out not only with new technologies but with high-end governance, ecosystem building, customer engagement and organizational capability of their platforms.

Organisations' priorities have also been changed by digital transformation. Digital transformation is about more than just adopting new technology, it is about rethinking the way business processes, organisational structures, leadership capabilities, and strategic decisions are made in an organisation. The digital state of readiness of organisations has thus

become one of the key factors in software commercialisation. Organisations with the right technological infrastructure, digital skills, leadership support, system of innovation and change management are more likely to be successful in commercialising their digital assets. Organisations with these are usually challenged to grow technically successful software products into commercially viable platforms, on the other hand.

APIs, or Application Programming Interfaces, are now key elements in the development of today's digital ecosystems, in addition to organisational capability. APIs can help platforms grow and expand by connecting with third parties, can help with innovation from third parties, and can help with interoperability between software systems. APIs that are designed well can be used to help developers use them, decrease integration complexity, and improve the participation of the ecosystem. API maturity is thus now emerging to become a strategic capability that helps platforms become more competitive and successful. However, API maturity is frequently assessed without considering other factors such as organisational and commercial factors.

Another area that has been gaining increasing research attention is platform governance. Governance mechanisms set the rules, standards, security requirements, quality assurance and participation policies for interactions in digital ecosystems. Good governance boosts the trust of the stakeholders, encourages participation of the ecosystem and facilitates sustainable development of the platform. Lack of good governance on the other hand can affect software quality, expand security risks, decrease third-party involvement, and ultimately have a negative effect on commercial sustainability. Thus, governance is an important linkage between software engineering and organisational management [20].

The technological and organisational changes have greatly contributed to the growth of existing knowledge but have also raised a number of research issues. The existing literature is scattered, and is organized by disciplines. The main research areas of software engineering are technical quality, software architecture, technical debt and maintainability. Information systems research is about preparedness, management, digital transformation and strategic capability of organisations. The Platform ecosystem research focuses on the analysis of

network effect, value co-creation and ecosystem participation. Business literature rates the performance of the business on financial and market-based indicators. There are few studies that take the different perspectives into account and provide a comprehensive view for assessing the readiness of commercial software during the software development process.

The fragmentation poses challenges for organisations that are trying to commercialise digital assets. Often decision makers look at single technical metrics, architectural analysis, or financial metrics when deciding about software investment opportunities. But commercial success does not occur on any one dimension, but through a combination of technical, organisational, governance, ecosystem and strategic dimensions. Currently, there are no all-encompassing decision-support frameworks that are able to objectively judge a commercial readiness of an organisation before a large-scale commercial deployment [21].

The above background and context highlight its growing complexity, and the need for an integrated evaluation framework to address it. In response to this need, this research offers a Commercial Viability Index (CVI) which integrates the technical, architectural, organisational, governance, ecosystem and commercial aspects into a single assessment model. The proposed framework is intended to overcome existing challenges to current evaluation practices to facilitate software developers, technology firms, investors and product managers to better inform their commercialisation decisions in moving from prototype to platform [22] [23].

To create a theoretical base, it is important that the key concepts of this research are understood. To transform software products from prototypes to a viable digital platform, several disciplines such as software engineering, information systems, digital transformation, platform economics and innovation management are required. Many of these ideas have been extensively studied separately but there may be different definitions of them in different fields of study. Thus, this section critically examines the key concepts that were used throughout this thesis, and define the operational concepts for the proposed Commercial Viability Index (CVI).

2.3 Definitions and Key Concepts:

For building up a consistent theory, it is crucial to understand the key concepts underlying this research. From prototype to commercially viable digital platforms, the implementation of software products requires numerous disciplines such as software engineering, information systems, digital transformation, platform economics, and innovation management. Many of these concepts have been studied separately but are not always clearly understood in the different fields of studies. For this reason, this part of the thesis reviews the key concepts which were used in all parts of this thesis and sets the operational definitions for the proposed Commercial Viability Index (CVI) [24].

2.3.1 Digital Asset

There have been significant changes in the definition of a digital asset over the last 20 years. Previous studies tend to characterize digital assets as electronic documents, multimedia content, databases and digital IP. But the evolution of cloud-based computing technologies, software engineering, and platform technologies has extended this definition to software applications, cloud-based services, APIs, machine learning models, digital platforms and more any technology-based resource that can create value for the organisation or business [25].

In software engineering, digital assets are emerging as a valuable resource for the organisation, not just a technological artefact. Digital assets are different from tangible assets in several ways, including their scalability, reusability, interoperability and evolution. These qualities make the digital asset an appealing option for commercialisation and help organisations scale to produce more digital products at a lower cost.

Various studies have suggested that digital assets have a utility larger than just for software. They become marketable if the software is of high quality, technologically adaptable, can be integrated in the ecosystem, is well-governed, is adopted by customers and is prepared by the organisations. Therefore, a technical-only analysis of digital assets is not enough to give a full picture of a digital asset's future trading value [26].

In the context of this research, a digital asset is considered to be:

A digital asset that is software-based, scalable as a platform, has an active ecosystem and drives business innovation to deliver sustainable organisational and commercial value, and continuously evolve technologically. This definition places a focus beyond the traditional software engineering concept by acknowledging that in order to be viable in the marketplace, software must be technically sound and capable of being run in an organisation.

2.3.2 Software Prototype

A software prototype is usually defined as an early working form of a software system that is built to demonstrate the technical viability of a software design, to validate user requirements or to assess a variety of software design options prior to a full-scale implementation.

Traditionally, prototypes were treated as a part of the development process that would be thrown away once the system has been validated. In today's software engineering, though, prototypes increasingly are seen as a first step in platform evolution. Prototypes have been evolving into continuously developed software products, which are capable of being commercialized without a redevelopment of the entire platform, through the use of agile development methodologies, DevOps practices and continuous delivery.

There are various types of software prototypes that can be distinguished by researchers:

- Throw-away prototypes
- Evolutionary prototypes
- Incremental prototypes
- Experimental prototypes

Of these, evolutionary prototypes are gaining significance as they are continually improved to become software systems suitable for production [27].

Even with technological advances, good prototypes often go down to failure in the commercialization stages. The literature reviewed indicates that the commercial sustainability cannot be guaranteed by technical feasibility. There are multiple factors that

affect the capability of a prototype to successfully transform into a digital platform such as, but not limited to, Architecture, Governance, Scalability, Organisational Capability, API maturity and ecosystem development.

Based on that, the following definition of software prototype has been made. Early operational software that serves as a platform for future platform evolution and commercialization and to validate technical functionality.

2.3.3 Digital Platform

One of the main topics of this research is the idea of the digital platform. Most of the early software systems were generally created to carry out a predefined organizational task. However, modern digital platforms offer a technological infrastructure for continuous interaction between multiple players, such as users, developers, organisations, partners as well as external service providers.

Digital platforms are unlike traditional software products in a number of ways. First platforms enable value co-creation and not value delivery. Second, platforms promote the innovation of third parties by providing them with APIs and software development kits (SDKs).

Third, platforms constantly change due to the participation in the ecosystem, not only through internal software development.

Examples include:

- Amazon Web Services
- Microsoft Azure
- Salesforce
- Shopify
- Google Android
- Apple iOS

These platforms are not just about the software, but about the ecosystem as well.

The researchers have found that a number of features of successful platforms have been identified consistently:

- Scalability
- Modularity
- Interoperability
- API accessibility
- Platform governance
- Network effects
- Ecosystem participation
- Continuous innovation

In this thesis digital platform is understood as:

A value creation software ecosystem that allows for interactions among key stakeholders, including through shared technological infrastructure, APIs, governance, and complementary innovation, that is scalable [28].

2.3.4 Platform Ecosystem

Platform ecosystems are the evolvement of the digital platform model considering the fact that software products don't always exist in a vacuum.

Rather, software relies more and more on interactions between:

- Developers
- Service providers
- Technology vendors
- Customers
- Regulators
- Business partners

According to platform ecosystem theory, the more successful business models rely on the development of ecosystems as opposed to software.

The more active the ecosystem and the more size it acquires, the more value can be brought to the platform by the businesses.

Several important ecosystem traits are identified:

- Network effects
- Third-party
- Collaborative value

development creation

•Complementary

innovation • API integration

- Knowledge sharing

Maturing ecosystems enhance software innovation and decrease an organisation's reliance on internal development resources. But the addition of ecosystems brings with it governance complexity. Platform owners have to set criteria for security, quality assurance, participation rules etc., API management, privacy and interoperability. This research thus regards the attractiveness of the ecosystem as an important dimension of the commercial viability [29] [30]

2.4 Theoretical Foundations

The digital asset from prototype to a viable, commercial digital platforms is a multidisciplinary issue and cannot be fully understood by a single theory lens. Commercialisation of software requires all these interdependent factors: technical prowess, organizational ability, ecosystem engagement, good governance, strategic flexibility and business value creation. Therefore, this study is based on a theoretical foundation that integrates Platform Ecosystem Theory, Resource Based View (RBV) and Dynamic Capabilities Theory. The same theories were also found in the research proposal as the conceptual underpinning for explaining the technical quality, organisation ability, involvement in the ecosystem and commercial viability [31].

These theories help to account for:

I. What are the factors that affect the success of software products in the marketplace (RBV); and

II. How organisations continually mould those capabilities in the fast-changing digital markets (Dynamic Capabilities); and

III. How external ecosystem participation and governance can change software products into scalable digital platforms (Platform Ecosystem Theory).

These theories don't contradict each other but rather supplement, providing explanations to various phases of software commercialization.

2.4.1 Resource-Based View

The Resource Based View (RBV) is a strategic management theory that is one of the most influential. RBV was developed by Werner felt (1984) and extended by Barney (1991) suggests that the key sources that are valuable, rare, inimitable and non-substitutable (VRIN) in an organisation are the sources that help them gain sustainable competitive advantage. Although competitive strategy is the traditional approach, which takes the external market positioning as the point of focus, RBV is a strategy which puts the emphasis on internal organisational resources as the point of focus for long term performance.

In software intensive organisations these resources are:

- Software architecture knowledge
- Organisational capabilities
- Development
- Source code quality
- Customer data
- Technical expertise
- Platform infrastructure
- Intellectual property

The RBV theory has been broadly utilized in previous research on digital transformation, software innovation, cloud computing and information systems. Research findings have always shown that those organisations with better technological resources tend to have more

innovation capability and better competitive performance. But, research in software engineering also indicates that it is not enough to have technical resources to be successful commercially [32].

For instance, a software prototype can have a good architecture, be technically debt-free and have advanced features and functionalities, and yet do not succeed in the market because of inadequate governance, participation of ecosystem or organisational capability. As a result, RBV cannot provide a full explanation as to why some digital assets have more commercial value, but it does offer some explanation about how organisations constantly transform these assets as platforms develop.

Digital markets are dynamic, and technological resources often become less competitive with the changing customer expectations and markets. RBV does not offer much guidance in terms of ongoing innovation or adaptation of the organisation [33].

However, the issues surrounding technical assets such as software architecture, APIs, organisational knowledge, and development capability, as key resources affecting commercial viability, are still very much relevant to this research and are addressed by RBV. In the proposed Commercial Viability Index, software architecture, API maturity and technical capability are strategic resources of the organisation in line with RBV.

2.4.2 Dynamic Capabilities Theory

RBV takes the idea further by illustrating the benefits of the internal resources of the organisation, but Dynamic Capabilities Theory further develops the idea to show how organisations continually adjust resources to the changing business environment.

Dynamic Capabilities Theory (DCT) was first proposed by Teece, Pisano and Shuen (1997) and states that in addition to having valuable resources an organisation must also be able to:

- sense new opportunities,
- seize emerging opportunities,
- transform existing capabilities,
- constantly change organisational resources.

In terms of the digital industries that feature swift technological evolution, short product lifecycles, and continuous innovation, these dynamic capabilities are especially significant [34].

Digital platforms don't stay the same in their technical sense. Software architectures change, APIs grow, governance matures, customers' needs change and ecosystems grow constantly.

As a result, software architecture will have to be redesigned constantly, governance will need to be enhanced, interoperability will need to be further developed and new platform services will need to be added to the mix.

Historical research on digital transformation always highlights the importance of how adaptable an organisation is in the long-term success of their business.

But there are restrictions on the Dynamic Capabilities Theory as well.

First, many studies have a high level of strategic conceptualization of dynamic capabilities and do not offer operational measures.

Secondly, empirical measurement is challenging as adaptability in organisations may be in the form of intangible managerial processes.

Third, there is disagreement about what should be considered as "dynamic capabilities" in the organization.

The theory serves to explain one very important reason why technically successful software prototypes often fail at the commercial level, even in spite of these limitations. However, businesses with low adaptability might find it challenging to adapt their architectures, reduce technical debt, improve governance or build platform ecosystems as market dynamics change.

So, the components of Dynamic Capabilities theory can be directly supported in the inclusion of:

- Digital organisational readiness
- Platform governance

- Continuous software evolution
- Organisational adaptability

In the proposed Commercial Viability Index.

2.4.3 Platform Ecosystem Theory

The Platform Ecosystem Theory is the most comprehensive theory from among all of the theories reviewed in this research to explain software commercialisation.

Value for traditional software products is generally created from direct transactions between software developers and customers.

Digital platforms are not like conventional platforms.

They become more dependent on the interplay of several groups of stakeholders such as:

- users, • technology partners, • complementors,
- developers, • service providers, • investor

According to the Platform Ecosystem Theory, the value of software platforms is derived from allowing these participants to engage in a series of interactions based on a shared technological infrastructure, not on software functionality [35].

Prior studies have shown that the following are necessary to enable successful platforms to become commercially sustainable:

- complementary innovation, • API integration,
- network effects, • collaborative value creation.
- third-party application

development,

The theory thus moves away from a focus on software products as discrete technologies to software ecosystems as innovation environments that are collaborative.

The following items from existing literature have been shown to be highly important for success of an ecosystem:

- governance mechanisms,
- API openness,
- interoperability,
- platform architecture,
- trust,
- ecosystem orchestration.

Research also suggests that in order for the platform to be viable, it needs to be able to attract and coordinate external party participation, not just meet internal software needs, according to the proposal. It also emphasizes the importance of governance, APIs and ecosystem orchestration in the path to platform success.

There are however, some caveats about Platform Ecosystem Theory.

First, in many situations studies of ecosystems are conducted on the basis that the software platform is already technically matured.

Second, there is a lot of ecosystem literature that ignores software engineering considerations like architecture quality, technical debt, maintainability, etc., as well as the implementation of the API.

Third, research on governance can be targeted at large and successful platforms, but less targeted at the early stages of software prototypes in transition to commercialization.

The restrictions of these call for the application of a Platform Ecosystem Theory and a software engineering point of view.

Here in this thesis ecosystem participation is seen as one aspect of commercial viability but not necessarily the only reason for commercial success.

Theoretical perspectives were integrated into the process of developing the lesson plan.

Each theory presents a particular aspect of the commercialisation of software.

The Resource Based View is an explanation of what valuable internal resources organisations have.

Dynamic Capabilities Theory describes how organisations are continually adjusting these resources.

Platform Ecosystem Theory describes how such capabilities create commercial value by engaging external parties and growing ecosystem.

The value creation is also highlighted to be not only based on the internal value creation but also on the interplay of technical quality, organizational capability, governance, APIs, and participation in the ecosystem which results in commercial viability [36].

Therefore, the present research suggests that the commercial viability arises when there is interaction between:

•Technical Resources (RBV)

The ability of an organisation to adapt and change is known as organisational adaptability or dynamic capabilities.

Platform Ecosystem Participation (Platform Ecosystem Theory)

- Governance
- API Maturity
- Software Architecture
- Technical Debt
- Digital Organisational Readiness

The proposed Commercial Viability Index (CVI) is based on this integrated perspective.

2.5 Existing Approaches and Technologies Review

This section looks at and reflects critically on the main technological solutions that have been suggested for enhancing the commercialisation of digital assets, following the structure suggested in the MR603 guideline. A literature review is not done in a linear fashion but instead is condensed into thematic areas showing common findings, contradictions, strengths, limitations and opportunities for further work. This will give a more solid

theoretical basis to the proposed Commercial Viability Index (CVI) and directly answer the research gap discussed in Chapter 1.

Modern digital platforms are not just technology, but a combination of technology capabilities. Current studies have shown that software architecture, technical debt management, API maturity, platform governance, organizational preparedness, cloud computing, and digital transformation strategies are all critical factors that contribute to successful platform commercialization. All these technologies have been widely studied and very few researchers have investigated the influence of these technologies on the commercial viability [37]

2.5.1 Software Architecture and Platform Scalability

Software architecture is known to be the backbone of software quality and system evolution. It outlines the overall architecture and design of a software system, such as the organization of its components, the ways in which they communicate with each other, the deployment strategies, and the design principles. A software architecture that is well designed can facilitate maintainability, scalability, interoperability, reliability, and continuous evolution of software [38]

In the past, enterprise software systems were mostly built with monolithic software architectures in which all functions were deployed as a single application. The software components of monolithic systems shared common resources and were fairly easy to deploy, making early development simpler. As organisations scaled their digital services, however, monolithic architectures were found to be very limiting.

Monolithic software systems have been shown to have several problems in research studies in a consistent manner:

- Limited scalability
- High deployment complexity
- Difficult maintenance
- Very tight cohesion in software elements
- Reduced development flexibility

Operational risk can rise when updating. Rise of operational risk during updates.

This restriction is especially relevant when software grows into a commercial online service that has a large number of thousands or millions of customers.

To address these issues, microservices architecture has been recommended by researchers more and more. In contrast with monolithic systems, microservices split software into autonomous services which communicate with one another via lightweight APIs. The services are independent and can be developed, deployed and maintained separately for added scalability, fault isolation and continuous software delivery.

Past empirical research shows that organizations that have embraced microservices tend to witness the following:

- Faster software deployment
- Improved system scalability
- Reduced development bottlenecks
- Greater organisational agility
- Better maintainability

The advantages of microservices are clearly seen in cloud-native platforms like Netflix, Amazon, Spotify, or Uber, which have proven to be effective in facilitating continuous platform evolution.

Even with these benefits it is acknowledged that there are a number of key issues.

Microservices introduce:

- Distributed system complexity
- Network communication overhead
- Service orchestration challenges
- Security concerns
- Increased monitoring requirements
- Complex debugging
- Higher governance requirements

Researchers have come to believe that while scalability is an important measure of architectural quality, it is also important to evaluate that quality in terms of governance capability, maintainability, interoperability, and organisational maturity.

This review thus proposes that the software architecture affects the commercial viability, not only by way of software functions, but also by the way software platforms evolve.

Technical debt is a one of the most researched subjects in Software Engineering.

2.5.2 Concept:

Maintenance costs that will arise in the future due to compromises in the design that were made in order to speed up the development of software. Some of these compromises can be the lack of documentation, duplicated code, not enough testing, temporary solutions of architecture and incomplete software refactoring [39].

Tech debt was first looked at as a matter of software quality that falls under technical manpower. Other, more recent studies say that technical debt is also an organisational and commercial issue as it reduces innovation potential, hikes up operational cost and hampers platform scalability.

As discussed in the literature there are numerous types of technical debt:

- Code debt
- Architectural debt
- Documentation debt
- Design debt
- Infrastructure debt
- Requirement debt
- Testing debt

Of these, architectural debt is the most important one as it has the most direct impact on software evolution.

The general consensus on the negative effects of unmanaged technical debt is:

- Reduced maintainability
- Increased development costs
- Slower feature delivery
- Lower software quality
- Reduced customer satisfaction
- Higher operational risk

But there are some other aspects of technical debt that are recognised in the literature, which are actually positive.

Strategic technical debt can speed up innovation by allowing for the swift creation of prototypes in extremely competitive market conditions, according to several researchers.

This viewpoint brings in a significant trade-off.

There could be some intentional technical debt which might help responsiveness in the market. Too much tech debt has a negative impact on long-term commercial sustainability.

Although there are many studies on technical debt in software engineering, there is limited research on the commercial results of technical debt. There are very few studies that focus on the effect of technical debt on:

- investor confidence,
- platform adoption,
- commercial scalability,
- organisational capability.
- ecosystem participation,

This is an important area which proposed Commercial Viability Index will fill. This goal includes three objectives: Version

2.5.3: API Maturity and Interoperability.

Application Programming Interfaces (APIs) have now come to play a vital role in software platforms today. APIs are becoming a key component in many digital platforms that enable

software services, third party applications, cloud infrastructure, business partners and external developers to communicate with each other.

API maturity is always regarded as a specific factor crucial for an ecosystem.

API maturity isn't just about the technology and equipment, it's also about:

- API documentation
- Security
- Reliability
- Version control
- Developer support
- Performance
- Governance
- Standardisation

Good APIs make things more interoperable, and allow for innovation from third parties. Many previous studies have shown that the organisations with well-designed APIs go through the following:

- stronger ecosystem participation,
- increased developer engagement,
- faster innovation,
- improved customer satisfaction,
- greater platform adoption.

In particular, the financial technology, healthcare, logistics, cloud and Software-as-a-Service sectors have seen an increase in importance regarding open APIs [40].

But, there are also some challenges to the API.

When it comes to API governance, these are some of the typical outcomes of poor governance:

- inconsistent documentation,
- security vulnerabilities,
- integration failures,

- developer frustration,
- ecosystem fragmentation.

API maturity frameworks in use today focus more on the technical aspects of the API and only consider the commercial aspects to a lesser extent. This means that there is a lack of maturity in the integration of API into commercial evaluation models.

2.5.4 Platform Governance

Platform governance is defined as the policy, standards and processes that are implemented to coordinate interactions across platform participants. Governance has become an ever-growing priority to ensure in the expanding digital ecosystem

- interoperability,
- ecosystem stability.
- trust,
- compliance,
- security,
- software quality,

Overall, there are two types of governance approaches. The first one is centralised governance, in which the platform owner holds a considerable amount of control over the software quality, API access, involvement of the ecosystem and the evolution of the platform.

The second fosters 'open governance,' opening up the innovation process to outside actors and a wider innovation ecosystem. Previous research shows that neither governance model is best on a global scale. Highly centralised governance has positive impacts on quality assurance and it might decrease ecosystem innovation [41].

There is a trade-off between openness of governance, innovation and security risks and platform consistency. Therefore, literature recently suggests balanced governance strategies, which include good quality control and flexibility of the ecosystem. Although much research has occurred, governance studies mainly focus on the more developed digital platforms. Few studies focus on governance in the early stages of software commercialisation, where the governance decisions can have a significant impact on the growth of the platform.

2.5.5 Digital Organisational Readiness

If only it were that easy to make a deal successful with technology. The capability of organizations is becoming more and more important, according to some researchers. Organizational readiness for digital means readiness of the organization to implement, manage and commercialize digital technologies.

Some common readiness dimensions are:

- Digital leadership
- Innovation culture
- Employee capability
- Strategic alignment
- Resource availability
- Change management
- Technology infrastructure

Up to now, several studies have consistently shown higher digital maturity of organisations is associated with better digital transformation results. But, the literature on organisations readiness does not often combine software engineering factors like:

- architecture,
- APIs,
- technical debt,
- platform governance.

To date, there is a limited understanding of software commercialisation as a result of this separation. This limitation is overcome by the proposed Commercial Viability Index which combines organisational readiness aspects with software engineering aspects [42]. The ability to transform your organisation using cloud computing. Transforming your organisation with

2.5.6 Cloud computing

Cloud computing has revolutionized the way software is being commercialised.

Cloud Infrastructure allows organisations to:

- deploy software globally,
- automate deployment,

- scale rapidly,
- improve resilience.
- reduce infrastructure investment,

Cloud-native architectures enable continuous integration, continuous deployment, and a quick platform evolution. As a result, cloud computing has emerged as an important key enabler of the digital platform business model. Likewise, the digital revolution is more than just about technology usage. Digital transformation is becoming more widely recognised as a shift in an organization that includes:

- business processes,
- organisational structure,
- customer engagement,
- innovation capability
- leadership,

To achieve digital transformation, it's important the technology is aligned with the organisation's strategy. The literature shows that organisations often spend a lot of money acquiring technology and they are not sufficiently aware about the organisational change management. This disparity can decrease the commercial success of good technical implementation [43]

2.6 Software Architecture and Digital Platform Evolution

A software system's architecture has been known for a long time to be one of the most significant factors in the quality, maintainability and long-term evolution of software. The complexity and inter-dependency of digital products are growing and decisions taken during architecture are impacting the performance of the product and the capacity of an organisation to successfully commercialise software. As a result, software architecture has become more than just a technical design consideration, it has become a strategic capability that plays a significant part in digital transformation, platform scaling, ecosystem building and commercial sustainability. Critically exploring the related literature on software architecture, as well as its role in the development and successful deployment of digital assets from

prototypes towards viable digital platforms. This section does not consider the architectural styles directly, but instead assesses the impact of architecture on scalability, maintainability, interoperability, organisational agility and commercial viability [44]

2.6.1 Evolution of Software Architecture.

Over the last 40 years, a lot has changed in software architecture. Primarily, monolithic architectures of software components were used in the early enterprise applications where all software components were running in a single executable environment. This made it easier to develop and deploy software modules as they all used the same database, programming language and runtime.

Monolithic architectures worked well with reasonably small software programs. But, as enterprises started to create more sophisticated enterprise systems, a number of drawbacks occurred. Monolithic systems have consistently been found to be complex to maintain, contain costs and become more resistant to technological changes as they get older. Simple updates to software can often be time-consuming to deploy, adding to the risk and slowing organisational agility.

The introduction of internet-based services, cloud computing and digital platforms that have been embraced by many organizations have led to greater demand for more flexible software architectures. This led to the gradual transition of software engineering towards modular architectures, service-oriented architecture (SOA) and more recently to microservices architecture. These approaches allow software to be broken down into small parts that can be changed without affecting other parts of the software system [45]

Throughout the literature, one can find many examples that architectural evolution is now a critical factor for organisations to continuously innovate their software. The architectures for modern software platforms must be able to accommodate regular software changes, independent deployments, quick scaling and continuous integration. As a result, the architecture has become more and more a means to business agility and not just a technical design option.

2.6.2 Monolithic Architecture

One of the most analysed architectural styles in software engineering is monolithic architecture. In monolithic systems, all the software applications are combined into one. The services and the supporting services, as well as the database, authentication mechanism, UI, and business logic, are all contained in a single deployment unit. Scientists list a number of benefits of monolithic systems.

These include:

- Simpler development processes
- Easier debugging
- Straightforward deployment
- Reduced communication overhead

Savings on initial development cost.

An easier time of testing early on in the development process. It is these properties that make many software start-ups start out with monolithic architectures when creating their minimum viable products (MVPs) and initial software prototypes. But literature consistently shows that there are a lot of disadvantages to growing systems.

Major limitations include:

- Limited scalability
- High maintenance cost
- Increased software complexity
- Longer deployment cycles

Integrated components in a tight coupling manner

- Reduced development flexibility

If software updates are made, there is an increased risk. There is an increased risk during software updates. When organisations grow their customer base, monolithic architectures can often become a barrier to business growth. As a result, numerous emerging technology firms over time chart a course towards more modular software architectures.

2.6.3 Microservices Architecture

Microservices architecture is one of the biggest advancements in modern software engineering. Compared to monolithic systems, microservices break up the functionality of

software into several different services. Services carry out certain business functions, interact with each other via lightweight APIs. There are many benefits of microservices that are discovered by the researchers.

These include:

- Independent deployment
- Improved scalability
- Greater fault isolation
- Better maintainability
- Continuous software delivery
- Technology flexibility
- Improved organisational agility

Netflix, Amazon, Spotify, Uber and Airbnb are all examples of cloud-native businesses that have proven the effectiveness of microservices for large software evolutions. The previous empirical research suggests that microservices organisations are able to deploy more frequently, have shorter software downtime, higher productivity, and higher platform scalability. But it is not as if there are not some significant omissions recognized in the literature.

Microservices increase:

- Distributed system complexity
- Network latency
- Service orchestration requirements
- Monitoring complexity
- Security challenges
- Infrastructure cost
- Operational requirements governance

Researchers are now increasingly focusing on how important it is for the organisation's maturity and governance capability to be present to ensure success of adopting microservices not just the architecture.

2.6.4 Cloud-Native Architecture

Cloud computing has changed a lot software architecture. Cloud-native architectures are architectures developed with the intent to leverage cloud infrastructure by using containerisation, orchestration technologies, automated deployment, elastic scalability and continuous integration. Cloud-native applications allocate compute resources on the fly, as demand rises and falls, in contrast to traditional enterprise software which is built to meet the needs of the organization at any given time. Cloud native architecture consistently has been linked to a few commercial advantages.

These include:

- Reduced infrastructure investment
- Improved disaster recovery
- Global scalability
- Continuous software delivery
- High availability
- Operational resilience
- Automated deployment

As a result, Software-as-a-Service (SaaS) platforms are now the most common applications using cloud-native architecture. However, cloud native systems need advanced governance, monitoring, cyber security and operational management features, too. This means that technological maturity is not enough for successful commercialisation.

2.6.5 This module covers the various software architecture quality attributes.

The architectural literature on Software Quality Attributes always mentions some of the quality attributes that affect software evolution. Some of the most common are:

a.
- Scalability
- Maintainability
- Reliability

ii.	Availability	v.	Modularity	viii.	Testability
iii.	Security	vi.	Interoperability	ix.	Flexibility
iv.	Performance	vii.	Portability		

Before the advent of this software engineering research, these attributes were typically each assessed separately.

The more recent studies acknowledge that they are interrelated in the software evolution.

For example:

I. Maintainability is often improved through improved modularity.
II. Technical debt can be minimized because of greater maintainability.
III. Technical Debt is reduced which helps continuous software evolution.
IV. Sustainability of the business is enhanced through continuous evolution.

This interaction implies that there are several interdependent links between software architecture and commercial results. The software should have a solid architecture, while maximizing its commercial viability. A central issue that has been topical in the literature recently is how software architecture relates to commercial success. Traditionally, architectural quality has been measured by technical performance measures such as execution efficiency, maintainability, scalability and reliability.

Recent studies to some extent suggest that architecture also affects:

I.	Product	VI.	Platform	X.	Organisational
II.	competitiveness	VII.	expansion	XI.	innovation
III.	Customer	VIII.	Ecosystem	XII.	Operational cost
IV.	satisfaction	IX.	participation	XIII.	Investment
V.	Time-to-market			XIV.	attractiveness

For instance, scalable architectures can help to lower infrastructure expenses and enable organisations to accommodate more customers.

Modular architectures make software improvement quicker, empowering organisations to adjust to the ebb and flow of the market.

Likewise, good APIs allow for greater ecosystem involvement with third-party innovation.

Thus, the ability to design architecture grows in both technical and commercial aspects too.

While software architecture is one of the most widely studied fields in software engineering research, there are still some key issues that have yet to be addressed.

Firstly, the majority of the architectural research concentrates primarily on the engineering outcomes.

Some of the key considerations to evaluating a location are:

i. Response time
ii. Software complexity
iii. Maintainability
iv. Deployment efficiency
v. Fault tolerance
vi. Scalability

There are relatively few studies assessing impacts of architectural decisions on commercial sustainability. Second, the topic of software architecture and organisational readiness do not appear to be combined in the software engineering literature.

The ability of the organisation, governance maturity, leadership support and technical expertise are critical factors for the successful implementation of the architecture. The dimensions are mostly lacking in conventional architectural studies. Third, numerous empirical studies focus on architecture in larger technology organizations that have large engineering staffs. Very little work has been done to date on start-ups or small organisations trying to commercialise early software prototypes. Last but not least, the literature on architecture often assesses the technical aspects only without considering issues of platform governance, API maturity, participation of the ecosystem, or business strategy. Thus, little guidance is currently available to inform the appropriateness of incorporating architectural quality into overall commercial readiness [46]

2.7 Critical Comparison and Synthesis of Previous Studies

Over the past decade, the body of work on digital asset commercialisation has grown significantly, and academic studies have focused on technical, organisational and business aspects that affect the process of scaling software prototypes to digital platforms. Most of the past studies focus on a single factor rather than linking it to a single framework for assessment of commercial viability. The majority of research is focused on one aspect of software architecture, platform ecosystems, technical debt, API management, digital transformation or organizational readiness, among others. So far, the research community has not developed an understanding of how these dimensions interact to create a platform that can succeed from the prototype stage to become a viable product.

The existing body of knowledge was critically assessed by comparing the existing studies most relevant to the present study in terms of research objective, methods, data used, conclusions, as well as the strengths and limitations of the studies and the relevance to the present study. It is important to note that this comparison reflects not only the level of maturity of knowledge, but also the gaps in knowledge that highlight the need for the proposed actions, specifically, the Commercial Viability Index (CVI).

Table 2.7.1 Critical Comparison of Previous Studies

Study	Problem	Method	Data/Sample	Main Findings	Strengths	Limitations	Relevance
Cusumano et al. (2019)	Platform business models	Case study	Digital platform firms	Ecosystems outperform standalone products	Strong industrial evidence	Limited technical evaluation	High

Baldwin & Woodard (2009)	Platform architecture	Conceptual analysis	Industrial cases	Modular architectures improve scalability	Foundational architecture theory	No quantitative validation	High
Bosch (2009)	Software ecosystem evolution	Industrial case studies	Software companies	Ecosystems improve innovation capability	Real-world evidence	Small sample size	High
Kruchten et al. (2012)	Technical debt	Literature review	Multiple software projects	Technical debt reduces long-term sustainability	Widely cited framework	No commercial metrics	High
Li et al. (2015)	Technical debt management	Systematic literature review	100+ studies	Early debt management improves maintainability	Comprehensive review	Lacks financial evaluation	High

Pautasso et al. (2008)	API design	Technical review	Web service systems	REST APIs enhance interoperability	Strong architectural guidance	Limited organisational perspective	Medium

Jacobson et al. (2012)	API maturity	Industrial practice	Enterprise software	Mature APIs accelerate platform adoption	Practical recommendations	Few empirical comparis ons	High
Teece (2018)	Dynamic capabilities	Theoretical analysis	Business organisations	Organisational capability drives competitive advantage	Strong strategic theory	Limited software-specific evidence	High
Bharadwaj et al. (2013)	Digital business strategy	Conceptual framework	Digital enterprises	Digital strategy integrates business and IT	Strategic perspective	Not platformfocused	High

Isckia (2009)	Open platform ecosystems	Case study	Apple App Store	Ecosystem openness increases innovation	Platform ecosystem insights	Single platform focus	Medium
Gawer & Cusumano (2014)	Industry platforms	Comparative case study	Technology firms	Platform leadership shapes ecosystem success	Strong comparative analysis	Limited software quality measures	High

Erdogmus et al. (2004)	Return on software investment	Economic modelling	Software projects	Software quality improves financial returns	Commercial perspective	Older industrial context	Medium
ISO/IEC 25010 (2011)	Software quality evaluation	International standard	Quality characteristics	Quality attributes affect product success	Standardised evaluation	No business indicators	High

Kitchenham et al. (2009)	Evidence-based software engineering	Systematic review	Software engineering literature	Rigorous evidence improves decision making	Strong methodology	No commercial framework	Medium
Fitzgerald & Stol (2017)	Continuous software engineering	Literature review	Agile organisations	Continuous delivery improves scalability	Modern development practices	Limited business outcome analysis	High
Blank & Dorf (2020)	Startup validation	Lean method ology	Startup case studies	Customer validation improves commercial success	Entrepreneurial perspective	Limited software architecture focus	Medium
Ries (2011)	Lean startup	Experimental iteration	Startup organisations	MVP testing reduces	Practical innovation framework	Limited platform maturity	Medium
				commercial risk		evaluation	
Parker et al. (2016)	Platform ecosystems	Business analysis	Global platform firms	Network effects determine platform success	Strong commercial insights	Limited technical assessment	High

Hein et al. (2020)	Digital platform ecosystems	Literature review	Platform ecosystem research	Governance and ecosystems jointly influence growth	Comprehensive synthesis	Limited quantitative validation	High

2.7.1 Critical Comparison of Previous Studies

As shown in the comparison, there has been a significant amount of research that has produced a substantial body of knowledge on successful digital platforms and its individual determinants. Yet, none of the studies currently available offers a complete framework that can assess the maturity level, organisational readiness, scalability of the platform, software quality, governance capability, API maturity, and commercial performance. Instead these dimensions are normally studied separately by separate research communities.

A major factor that stands out from the literature is that the importance of software architecture as a key factor influencing platform scalability is consistently mentioned. All three—modularity, maintainability, interoperability, reliability and scalability—are all found by Baldwin and Woodard (2009), Bosch (2009) and ISO/IEC 25010 to be essential qualities of successful software platforms. In the same way, the research on technological debt conducted by Kruchten et al. (2012) and Li et al. (2015) shows that bad architectural choices at the prototype stage can have a substantial impact on the commercial potential in the future, due to higher maintenance expenses and lower speeds of innovation. The studies all point to technical quality as one of the best indicators for long-term sustainability of platforms.

There is another common thread in terms of platform ecosystems and governance. Tawana (2014), Gawer and Cusumano (2014), Cusumano et al. (2019), Parker et al. (2016), and Hein et al. (2020) all point out that the governance structure, involvement of ecosystem members, and network effects are all key factors in the success of a digital platform, in addition to technical excellence. These studies provide evidence that being successful is no longer just about providing a high-quality software product for the platform, it's about attracting complementary developers, partners and users to it as well. However, the majority of ecosystem research is conducted on

established platforms in the global market – Apple, Amazon, Google, and Microsoft – and little has been done on early-stage digital assets and prototype commercialisation.

The results of studies on organisational capacities are also quite consistent. Teece (2018) and Bharadwaj et al. (2013) state that organizational readiness, leadership ability, strategic fit and digital transformation are crucial factors affecting commercial success. These results indicate that platform success is not only about technical skills; there is also a need for a platform's evolution to go hand in hand with organizational capabilities. However, most of these studies come from the strategic management field and do not have detailed software engineering metrics as part of their analyses.

The review also reveals that the area of API maturity has been gradually emerging as a significant area of research. Pautasso et al. (2008) and Jacobson et al. (2012) end by saying that these are the benefits of mature APIs – improving interoperability, enabling third party innovation and spurring ecosystem growth. API quality is directly tied to the potential digital assets have to grow and scale, allowing external integration and expansion of services. Yet, API maturity is not well understood or integrated in the commercial evaluation models.

Conceptually, there is a prevalent body of knowledge of methodologically systematic literature reviews and conceptual frameworks. Kitchenham et al. (2009), Li et al. (2015), Fitzgerald and Stol (2017), and Hein et al. (2020) use sound evidence synthesis techniques with wide scope of theory building. In the meantime, a number of industrial practice studies have been carried out by Bosch (2009), Cusumano et al. (2019) and Isckia (2009) provide some useful practical insights into platform evolution. These methodologies result in good conceptual understanding, but to date few studies have created predictive models that can quantify commercial viability.

Case studies appear to be one of the most widely used research techniques in the literature, as they enable a researcher to examine the interaction of a number of technological and organisational components in an actual industrial context. These studies are however often limited to a smaller number of organisations that limits generalisability of results. On the other hand, systematic reviews offer more compelling evidence since they pool together several studies, but they are not able to establish any causal links between the technical characteristics and commercial results.

There is also some consistency in the literatures with regard to the data sets and research contexts used. Empirical research typically deals with software companies, platform providers, technology start-ups, enterprise software vendors or and large digital ecosystems, mostly in North America and Europe. The most common datasets are open-source software repositories, industrial software projects and commercial platform case studies. In comparison, there is little evidence related to small and medium enterprises (SMEs), Australian software organisations, university developed prototypes, government digital assets or early stage research commercialisation projects.

A number of methodological flaws emerge time and time again in the past. Firstly, many investigations focus on a single dimension of platform success – software quality, governance, software debt or organisational capability – and ignore interactions between these dimensions. Second, cross section study designs are prevalent in literature and it is difficult to know how digital assets change over time. Third, there are few studies that use data sets that enable the identification of prototype transformation into commercial platforms. Fourth, there is limited quantitative validation and very few proposed frameworks have been statistically tested in a variety of industrial settings. Last but not least, many conceptual models lack measurable indicators for commercial readiness which can be directly used by practitioners when they assess the commercial readiness.

While there is consensus on the significance of software quality, capability of organisations, governance, and platform ecosystems, there are some conflicting results. The principles underlying the Lean Start-up (Blank & Dorf, 2020, Ries, 2011) suggest, for instance, that commercial validation via the market should happen quickly with minimum viable products, rather than technically focus on perfection. For software engineering, by contrast, poor software design and technical debt are said to have outstanding long-term effects, such as high maintenance costs in the future that might arise from premature commercialization. Likewise, in the field of ecosystem research, open platform strategies are often encouraged to foster innovation, while in the field of governance research, it is stressed that there is a need to introduce a level of control of the platform to guarantee quality and security. The opposing views point to a need to balance the technical maturity, organisational readiness and market responsiveness of a product when trying to successfully commercialise it, not just focus on one.

In general, it shows that there is a considerable body of knowledge on the existing literature on various determinants of digital platform success, but there is a gap in research that could provide a holistic framework to measure the overall commercial viability of digital assets. This is the main motivation for the research presented here. This thesis introduces a new framework for measuring the overall commercial viability, the Commercial Viability Index (CVI), which brings together software architecture quality, technical debt, API maturity, platform governance, organisational readiness, scalability and commercial performance into a single quantitative assessment framework, unlike previous studies. This holistic approach represents a huge gap in literature and provides both theoretical progress as well as practical support for decision making in the process of evolving a software prototype into a commercially viable digital platform.

2.8 Challenges, Limitations and Unresolved Issues

While there has been considerable research and development in both software engineering and digital transformation, there are still several challenges to understand how software prototypes can successfully transform to commercially viable digital platforms and what factors contribute to this. Previous work can be useful for technical and/or business aspects, but few integrate these aspects in a comprehensive framework of commercial viability assessment. The proposed Commercial Viability Index (CVI) in this research is because of the unresolved issues.

One of the biggest constraints is the lack of the adequate and imbalanced data sets. Most studies are based on case studies of large technology firms (Bosch, 2009; Cusumano et al., 2019) or on open-source repositories or proprietary organisational data. The datasets are frequently limited to just successful products, start-ups and small organisations, which reduces the representativeness of research results. In addition, data collections do not usually include technical software metrics and business performance measures, making it hard to evaluate them together in the commercial market.

Consideration of privacy and security issues are also major challenges. Organisations do not like to provide sensitive information on software architecture, financial performance, customer usage, operational processes etc. because of confidentiality and regulatory requirements (Bharadwaj et al. 2013). This has made it difficult to have access to an extensive real-world dataset, thus limiting the opportunities of empirical validation.

The other one is algorithmic bias, which hasn't been resolved yet. Artificial intelligence and machine learning models heavily rely on past data which may not be representative of various industries, sizes of organisations, or geographical areas (Mehrabi et al., 2021). As a result, predictive-models might have a bias for large tech companies and yield less accurate results for other start-ups, SMEs or digital assets developed at universities.

Another issue recognized in the literature is the lack of explanation of sophisticated predictive models. While machine learning approaches can yield prediction accuracy, many can be considered as a "black-box" with their decision-making process hard to interpret (Rudin, 2019). This opacity diminishes the trust between managers related to investment and commercialisation decisions.

The other limitation is the lack of generalisability. The bulk of empirical research is in relation to global platform organisations like Microsoft, Google, Apple and Amazon (Gawer & Cusumano, 2014). Therefore, the information that is already known may not be directly transferable to research-based digital innovations or to smaller organisations or government projects. The limited Australian evidence is also important to highlight, with relatively few studies focusing on Australian software companies, or digital innovation ecosystems.

However, a few studies do have high computation costs, especially for analyzing large software repositories, source code metrics, and data from the platform ecosystem. Such computational needs might make the implementation of the technique impractical, particularly for organisations with limited technical resources.

There is a lack of evaluation in the real world, which is a common problem in the literature. Most of the proposed frameworks are evaluated by simulation or historical data instead of software projects that are in operation. Again, small sample sizes decrease the statistical reliability, and weak validation procedures can only be used to assess one data set and may not be independent, with comparisons to other industry data sets.

One of the other key areas of unmet need is a lack of longitudinal studies. Most studies rely on cross sectional data that are collected at one point in time, but software commercialization is a long-term process with changes in software architecture, organizational capability and market

uptake. There have been few long-term studies of the evolution of prototypes to successful platforms.

The review further shows that there are no comparisons with baseline methods. Many researchers do not compare their new commercialisation models or prediction models with other software maturity models or with other traditional statistical models. It is not easy to tell if the suggested techniques will offer substantial gains without comparisons against a standard.

Last but not least, ethical and regulatory concerns are neglected in existing research. Organisational information, customer behaviour and Artificial Intelligence are increasingly being used for commercial viability assessments, causing concerns about privacy, transparency, fairness and regulatory compliance. However, commercial evaluation frameworks typically do not include standards like the ISO/IEC 25010, ISO/IEC 27001, GDPR or the Australian Privacy Act.

So far, literature has shown that while there is great progress in the understanding of software quality, platform ecosystems and digital transformation, there are still plenty of gaps to address with further research. Most of the works made before focus only on technical, organisational or commercial aspects separately. Known limitations such as lack of data, privacy concerns, algorithmic bias, weak explanation, inadequate validation, etc., continue to hinder practical application of the algorithm, as does lack of evaluation in the real world, limited Australian evidence, and lack of longitudinal studies. The main driving force for this research is the lack of an overall and evidence-based tool to measure the commercial readiness of digital assets moving from prototype to scalable digital platforms, where the desire to create a Commercial Viability Index (CVI) is the primary focus.

2.9 Research Gap

While there has been considerable research and development in both software engineering and digital transformation, there are still several challenges to understand how software prototypes can successfully transform to commercially viable digital platforms and what factors contribute to this. Previous work can be useful for technical and/or business aspects, but few integrate these aspects in a comprehensive framework of commercial viability assessment. The proposed Commercial Viability Index (CVI) in this research is because of the unresolved issues.

One of the biggest constraints is the lack of the adequate and imbalanced data sets. Most studies are based on case studies of large technology firms (Bosch, 2009; Cusumano et al., 2019) or on open-source repositories or proprietary organisational data. The datasets are frequently limited to just successful products, start-ups and small organisations, which reduces the representativeness of research results. In addition, data collections do not usually include technical software metrics and business performance measures, making it hard to evaluate them together in the commercial market. Consideration of privacy and security issues are also major challenges. Organisations do not like to provide sensitive information on software architecture, financial performance, customer usage, operational processes etc. because of confidentiality and regulatory requirements (Bharadwaj et al. 2013). This has made it difficult to have access to an extensive real-world dataset, thus limiting the opportunities of empirical validation.

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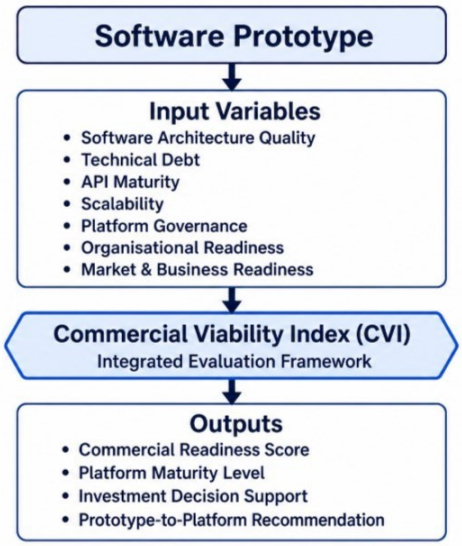
Gap Type	Gap Identified	Research Contribution
Knowledge Gap	Limited understanding of overall commercial viability.	Develops an integrated Commercial Viability Index (CVI).
Methodological Gap	Existing studies are mainly qualitative or conceptual.	Proposes a quantitative evaluation framework.
Technical Gap	Technical factors are evaluated separately.	Integrates multiple technical indicators into one model.
Validation Gap	Limited real-world and comparative validation.	Validates the framework using empirical evidence.
Contextual Gap	Limited Australian evidence.	Provides findings relevant to Australian organisations.
Practical Gap	Lack of practical decision-support tools.	Develops an industry-oriented commercial viability assessment framework.

Summary of Research Gaps

2.10 Proposed Conceptual Framework (Research Positioning)

The literature review revealed that there are certain research gaps such as the absence of an integrated framework for assessing the commercial viability of digital assets. Previous research is limited to each of software quality, software architecture, platform governance, technical debt, API maturity, organisational readiness and commercial performance. But none of these dimensions has been combined in one framework to evaluate the readiness of a software prototype for scaling up to a digital platform. To overcome this drawback a commercial viability index (CVI) has been suggested in this research as an all-in-one evaluation model.

The conceptual framework proposed brings together the technical, organisational and commercial aspects, into one evaluation model. It assumes that, in addition to software quality, the ability to make the software commercially viable relies on the readiness of the platform, the governance, as well as the performance of the market. The framework offers a systematic way to capture the readiness of a commercial product and the evidence to support decision making.



Proposed Conceptual Framework

The framework states that enhancements to the software architecture, API maturity, scalability, governance and organisational readiness will have a positive impact on the Commercial Viability Index. The higher the CVI, the more likely a digital asset is to succeed in becoming a commercially viable platform.

The following research propositions are developed with the basis of the proposed framework:

P1: The Commercial Viability Index is positively affected by Higher Software Architecture quality.

P2: Technical debt has a negative impact on commercial viability.

P3: The further refinement of APIs makes platforms readier for commercialization.

P4: Business success is supported by having a good level of organisational readiness.

P5: Effective platform governance facilitates platform to prototype.

P6: Digital assets are more commercially viable the more scalable they are.

P7: The Commercial Viability Index is positively impacted by greater market and business preparedness in.

The proposed framework is new as it combines technical, organisational and commercial aspects in a single evaluation framework. Previous research tended to deal with either a single aspect of platform development (e.g., software quality) or with a single business strategy (e.g., business strategy). This research, however, takes several factors into account to give a more comprehensive evaluation of commercial readiness.

Moreover, the framework is conceived of as a model of decision making and not implementation. It has been developed to help software developers, researchers, start-up founders, investors and technology managers determine if a digital asset is sufficiently mature to be commercialized. Overall, the suggested Commercial Viability Index (CVI) offers a new research tool that overcomes the gaps in knowledge, methodology, technique and practice found in the literature. The framework encompasses a multi-dimensional view of the software and business readiness, providing a systematic and evidence-based way to assess the commercial viability of digital assets and enable their successful commercialization from a prototype to a scalable, digital platform.

2.11 Novelty and Research Contribution

The literature review shows that the research field of digital asset commercialisation is still not coherent across the fields of software engineering, digital transformation, platform ecosystems and business management. There have been a few individual studies on software architecture, technical debt, API maturity, platform governance, organisational readiness and commercial performance. There has been, however, limited research that considers these dimensions as a whole and assesses the readiness of a software prototype to be a commercially successful digital platform.

The six key dimensions covered in this research—software architecture quality, technical debt, API maturity, platform governance, organisational readiness and commercial readiness have been brought together into a single framework, the Commercial Viability Index (CVI), which presents a novel approach to the research. The proposed CVI has advantages over other models like

Technology Readiness Level (TRL), Market Readiness Level (MRL) and Capability Maturity Models as it evaluates technical and organisational readiness for commercialisation at a holistic level.

One of the other key contributions is the emphasis on the feasibility assessment of early-stage commercialization. While there are existing studies that assess the performance of software once it has been deployed in the market, the primary measures are financial or customer adoption metrics. The proposed CVI, on the other hand, is used to evaluate the commercial readiness in the prototype and development phases, which allows the organisations to identify the technical and organisational weaknesses before they commit to a big investment. This proactive measure can decrease the commercial risk and make better decisions in software development.

Theoretically, this study combines three theories: Resource Based View (RBV), Dynamic Capabilities Theory and Platform Ecosystem Theory. Previous research has adopted these theories individually, but this research integrates these theories to understand how they all interact in determining commercial viability.

The study also delivers practical implications by providing a decision support for software developers, technology start-ups, product managers, investors, and innovation leaders. The proposed CVI can help make better commercialisation decisions by assessing readiness from various angles in a single framework, instead of just software quality or financial aspects.

In general, the primary contribution of this research is the integration. In contrast to the above-mentioned studies, which mostly focused on the technical quality of the software, the current research aims to answer the question: is the software technically, organisationally and commercially ready to be a successful digital platform? This research fills an important literature gap by combining a technical and organizational perspective with a commercial viewpoint - in one Commercial Viability Index - and contributes theoretical and practical insight into the research of digital asset commercialisation.

2.12 Chapter Summary

This chapter critically examined the literature concerning the commercialisation of digital assets and how these have developed from software prototypes to the creation of a scalable digital platform. The review covered the theory, concepts, and current studies of software architecture,

software technical debt, API maturity, platform governance, digital organisational readiness, cloud computing and platform ecosystems. It also examined major theories that support this study, which include the Resource-Based View (RBV), Dynamic Capabilities Theory and Platform Ecosystem Theory, all forming the theoretical basis for the understanding of commercial viability.

Numerous studies demonstrate that software quality, scalability, ability to develop APIs that are mature, good governance, and organisational readiness are critical to successful platform development. The factors have been explored in most previous research using either conceptual frameworks, systematic literature review, or industrial case studies. These strategies have proved useful in contributing to the knowledge of the individual technical and organisational aspects but have largely analysed them separately from the other factors influencing commercial success.

Some important limitations of the existing literature were also provided in the review. The state of the art is still scattered throughout the software engineering, information systems and business research fields, and a comprehensive framework for assessing commercial viability does not exist. Most studies have used small samples, cross-sectional studies, and have not been well tested against reality in the real world in this respect. In addition, few studies have addressed the particular case of organisations, start-ups or early stage software prototypes from Australia, limiting the usefulness and generalisability of previous research.

Based on the aforementioned results, the chapter concluded that there is a definite research gap. While there are several studies that have explored software architecture, technical debt, API maturity, governance, organisational readiness and commercial performance, there is no framework that integrates all these aspects and determines whether the software prototype is commercially mature to be further scaled to be a digital platform. This gap underscores the importance of a multi-disciplinary evaluation strategy which considers both technical and organisational aspects.

To fill this gap, this research introduces a Commercial Viability Index (CVI) which is derived from software architecture quality, technical debt, API maturity, platform governance, organisational readiness, scalability, and commercial readiness, into one assessment framework. The conceptual framework and research positioning outlined in this chapter lay the foundation of the research and its potential academic and real contributions.

The methodology followed to develop and validate the proposed Commercial Viability Index is discussed in the next chapter. It outlines the research philosophy, research design, data collection, analysis, model building, and model validation approaches to be adopted in the evaluation of the commercial viability of digital assets from prototype to large scale digital platforms

Chapter 3: Research Design, Methodology and Requirements

3.1 Chapter Introduction

The operationalization of the research direction set in the Chapter 1 and 2 is presented in this chapter. The background, problem statement or literature synthesis is not repeated but rather the development and empirical examination of the proposed Commercial Viability Index (CVI) is described. The design includes the approved constructs (software architecture quality, technical debt, commercial performance, API maturity, platform governance, digital transformation capability, organizational readiness), as measurable variables and an explicit analysis procedure.

The study is set in an empirical framework-development and validation study with an overall quantitative approach, through a pragmatic methodological approach. The survey data available includes 250 complete responses and 28 construct indicators and further codes provide details on the role of the respondent and the type of organization. The IBM SPSS Statistics software was proposed for descriptive statistics and preliminary screening, whereas the SmartPLS 4 software was proposed for the assessment of the measurement model and structural model of PLS-SEM. The present chapter also introduces the concepts of sampling, data preparation, data validation, ethical controls, requirements, and procedures to guarantee the reproducibility of the results. One of the key quality principles is ‘traceability' – every research question is associated with an objective, evidence source, analytical procedure and expected output. This helps to avoid the methodology to become a list of statistical methods and helps to ensure that the proposed CVI is directly supported by the empirical evidence [47] [48].

3.2 Research Design and Methodological Positioning

The research takes a pragmatist stance as the central problem is both explanatory and applied; the research first has to identify which dimensions of the three pillars – technical, organizational,

and commercial – are associated with commercial viability and then combine these dimensions into an applicable assessment framework. There is a place for pragmatism when the type of method used is determined by the research problem and/or the value of the evidence produced by the method and not by the allegiance to a particular philosophical tradition. The design of the operation is a framework-development and empirical-validation design. The quantitative part is the major evidential part as hypotheses involve the estimation of relationships between the latent constructs. As this study is prediction- and frameworkoriented, it includes multiple latent constructs that are measured by several indicators, as well as aims at estimating a series of simultaneous relationships instead of a series of disconnected bivariate tests, PLS-SEM is chosen as the main structural modelling technique. Diagnostic and explanatory analyses are descriptive statistics, reliability analysis and correlation analysis which are used as complementary to the structural model, but not as substitutes [49] [50]

The overall project logic for these explanatory sequential mixed methods is outlined in the previous research plan: quantitative data determine the strength and direction of the proposed relationships; then, expert evaluation can be used to interpret practical relevance, to improve the CVI weighting and to evaluate usability. But only evidence available in the given data set of 250 responses is reported in Chapter 4. It thus does not act as a substitute for the collection of expert evidence and does not introduce evidence as if it had been collected.

3.3. Research approach and method selection

The quantitative analysis is structured in the four levels. The data is first examined for completeness, valid range, duplicate data and coding consistency. Secondly, descriptive statistics at an item and construct level are generated to gain knowledge of central tendency and dispersion. Thirdly, there is an analysis on internal consistency reliability and construct relation. Thirdly, the proposed PLS-SEM model is evaluated in SmartPLS 4 based on the measurement-model and structural-model. This staged approach provides the separation of data quality, measurement quality and hypothesis testing.

The advantage of using PLS-SEM over using multiple regression is that instead of observed single variables, the CVI is represented by multidimensional constructs. Regression may be helpful as a check of sensitivity for a diagnosis, but a thesis statement about their construct effects should be made based on the validated measurement model and the structural model. Therefore, in the

final analysis indicator loadings, internal-consistency reliability, convergent validity, discriminant validity, collinearity, path coefficients, explanatory power and effect size and predictive relevance of indicators should be reported before drawing conclusions about the hypotheses [51]

Method	**Purpose in this study**	**Primary output**
Data screening	Establish data integrity and analysis readiness	Quality log and retained sample
Descriptive statistics	Profile respondents, items and construct scores	Frequencies, means, SDs
Cronbach's alpha / composite reliability	Assess internal consistency	Reliability evidence
PLS-SEM measurement model	Assess indicator and construct measurement quality	Loadings, CR, AVE, HTMT
PLS-SEM structural model	Test H1–H6 and explanatory/predictive relationships	β, t, p, CI, R^2, f^2, Q^2
CVI construction	Integrate validated dimensions into a readiness score	CVI score/matrix
Expert evaluation (planned)	Assess interpretability, applicability and comparative usefulness	Practical validation for RO5/H7

3.4 Alignment of Research Questions, Objectives, Methods and Evidence

Research question	Aligned objective(s)	Evidence/method	Analysis/evaluation	Expected output
RQ1: significant technical, organisational and commercial factors	RO1–RO2	Literature-derived constructs + 250-response survey	Descriptive profile; measurement validation	Validated factor set
RQ2: interaction among the principal constructs	RO3	SAQ, TDM, APIM, OR, PG, DTC and CPI indicators	Correlations and PLS-SEM structural paths	Direction and strength of relationships
RQ3: integration into the CVI	RO4	Validated construct scores and structural evidence	Weighting/normalisation and index design	Operational CVI matrix
RQ4: effectiveness relative to isolated approaches	RO5–RO6	Empirical model + planned expert/baseline comparison	Reliability, predictive performance and comparative evaluation	Evidence of applicability and limits

It shows that there is no research question answered by a method that is not related to its evidential requirement. The main type of question for RQ1 is a construct-identification and measurement question, for RQ2, it is a relational question, for RQ3 it is an artefactdevelopment question, and for RQ4, it is a validation question that is not simply a correlation question. Specifically, the provided survey can be used to provide preliminary evidence for RQ1-RQ3, and a defensible response to RQ4 will be completed through the planned comparative and expert-evaluation activities.

3.5 Research Context, Research Data Sources and Participants

For the present stage, a structured survey with 250 respondents is used as an empirical source. For each record a unique respondent identifier is included, along with three coded profile variables (Role, Experience and Organization Type), and 28 Likert style measurement items. There are seven constructs, each consisting of four measurement items: SAQ, TDM, APIM, OR, PG, DTC and CPI. The response range provided for the file is 1-5.

Construct	Code	Indicators	Role in model
Software Architecture Quality (SAQ)	SAQ	SAQ1, SAQ2, SAQ3, SAQ4	Predictor
Technical Debt Management (TDM)	TDM	TDM1, TDM2, TDM3, TDM4	Predictor
API Maturity (APIM)	APIM	APIM1, APIM2, APIM3, APIM4	Predictor
Organisational Readiness (OR)	OR	OR1, OR2, OR3, OR4	Predictor
Platform Governance (PG)	PG	PG1, PG2, PG3, PG4	Predictor
Digital Transformation Capability (DTC)	DTC	DTC1, DTC2, DTC3, DTC4	Predictor
Commercial Performance Indicator (CPI)	CPI	CPI1, CPI2, CPI3, CPI4	Commercial outcome

The dataset is directly linked to the conceptual framework presented in the 1st and 2nd chapters as each major construct has four observable indicators. Commercial performance is reflected in CPI1–CPI4, and serves as the main empirical result of the preliminary relational investigation. It is important to note that, when the "items" have a greater debt burden the direction of the technical debt should be interpreted as inverse, similar to the negative hypothesis of the commercial performance: higher item values in the supplied data should correspond to lower TDM. The exact wording of the questionnaire should be maintained in the final thesis in an appendix so that this direction can be verified unambiguously.

3.5.1 Online PLS-SEM Benchmark and Documentation Sources

The study is also informed by official PLS-SEM documentation and sample-project repositories to boost methodological reproducibility and offer recognized external benchmarks for the PLS-SEM workflow in SmartPLS. The SmartPLS Sample Projects repository contains downloadable benchmark projects and data files for various applications of PLS-SEM; the PLSSEM companion website contains textbook datasets, SmartPLS project files and case-study resources for established PLS-SEM guidance. They are not an alternative to the empirical data (250 responses) collected from the study itself but instead serve as methodological and software validation references [52]

The European Customer Satisfaction Index (ECSI) sample project is one that is useful as a benchmark: SmartPLS reports a sample of 250 observations from the French mobile-phone industry and serves as the model for PLS-SEM and related SEM approaches. The ECSI project can be employed to validate the model building, importation, estimation and reporting process in the SmartPLS 4. The ECSI observations and indicators are not presented in this thesis as the source of the following variables for the analysis of the SAQ, TDM, APIM, OR, PG, DTC and CPI: however, this does not mean that the data presented in the thesis is not from the survey; the survey administration data must be documented from the actual survey administration records [52]

Online resources: SmartPLS Sample Projects; PLS-SEM Companion Downloads; SmartPLS ECSI Benchmark Model.

3.6 Sampling and Data-Selection Strategy

The analysis is based on all 250 given responses and not a secondary subsample. The inclusion rule is a record that contains a valid unique Resp_ID and valid values for the profile variables and for all 28 construct indicators. This allowed the initial screening to identify 250 unique respondent identifiers, no missing values, and no exact duplicate rows suggesting no responses were dropped at this stage. This is a full case analysis of the available dataset which does not lose any information [53].

The recruitment process for the final thesis should be described precisely including the population to which the results of the recruitment process are generalizable, the recruitment process (convenience sampling or purposive sampling or snowball sampling), and the inclusion criteria. The numerical data does not provide enough information to get this information. These should be obtained from the record of the survey administration and the ethics papers. Sample size is important because it is sufficient for the initial multivariate analysis; representativeness relies on the process of sample selection, not the number of responses.

Profile variable	**Code distribution in supplied data**	**Total**
Role	1: 67; 2: 66; 3: 44; 4: 40; 5: 33	250
Experience	1: 59; 2: 86; 3: 72; 4: 33	250
Organisation type	1: 99; 2: 81; 3: 38; 4: 32	250

3.7 Research Instrument and Data Collection

A questionnaire is used to operationalize each latent construct and has four items. The multiitem structure is preferred over a single item score, in that it provides an opportunity to evaluate the measurement reliability and validity prior to the interpretation of the structural relations. The indicators all have 5-point scales in the provided dataset. The profile variables should be coded numerically and a profile coding book that specifies the substantive meaning of each code should be included in the final appendix. It should not be a spreadsheet but rather a research instrument that is documented via the questionnaire. The last submission should contain the items wording, construct source/adaptation rationale, response anchors, statement of the information for the participants, consent process, estimated time of completion and any pilot revisions. If they were adapted from previous literature, the original literature should be referenced on the measurement table. If it is a researcher developed item, the rationale based on theory expert pilot review should be reported

Instrument component	Measures	Quality control	Related RQ
SAQ1–SAQ4	Architecture quality	Item screening, reliability, convergent/discriminant validity	RQ1–RQ3
TDM1–TDM4	Technical debt / debt management dimension	Direction check, reliability, validity	RQ1–RQ3
APIM1–APIM4	API maturity	Reliability and validity	RQ1–RQ3
OR1–OR4	Organisational readiness	Reliability and validity	RQ1–RQ3
PG1–PG4	Platform governance	Reliability and validity	RQ1–RQ3
DTC1–DTC4	Digital transformation capability	Reliability and validity	RQ1–RQ3
CPI1–CPI4	Commercial performance	Reliability, validity and endogenous-model assessment	RQ1–RQ4

3.8 Proposed Commercial Viability Index Framework

The proposed CVI is not a "sum of the scores" of the items on the questionnaires. It takes a measurement first approach to development. The indicators of each construct are first presented, followed by evaluation of reliability and validity. Constructs with poor measurement properties

should not be included in the total index. The structural model then makes predictions about the relationship between the predictor dimensions and the commercial performance and offers empirical support to the direction and relative importance of the dimensions.

There are three conceptual layers in the framework. The 28 indicators from the survey are in the input layer. SAQ, TDM, APIM, OR, PG, DTC and CPI are part of the construct layer. The decision layer uses validated construct data to generate an interpretation of the level of commercial readiness. Positive readiness dimensions are expected to make the CVI higher and technical debt will make the CVI lower. The final weighting method needs to be determined after the measurements and after structural validation and should be assessed using a sensitivity analysis instead of just being selected to best fit this sample.

In general, a CVI can be written as CVI = Σ (wj × Zj), where Zj is a normalized validated construct score, and wj is the justifiable weight of that score. If the debt construct, then the orientation should be reversed or the weight should be negative, otherwise, the readiness level will be artificially increased by debt. Weights can come from a validated structural evidence, expert judgement, or an explicit hybrid. The final thesis needs to show how the chosen weighting system works compared to an equal weight baseline to illustrate that the proposed CVI has a positive contribution to make over simple aggregation.

3.9 Data preparation and processing

Data preparation was predetermined prior to substantive interpretation. The file provided was converted to a comma separated table. Resp_ID was considered as an identifier and not as an analytical predictor. The coded profile variables included were: Role and Exp and Org_Type. The 28 construct indicators were assessed for missing items and range of responses. There were no missing values or duplicate rows found, and the range of values for the indicators were as expected 1–5.

Arithmetic means were used to represent each construct (4 items) for preliminary construct level exploration. In Chapter 4 these means will only be utilized for descriptive and diagnostic analyses. The last PLS-SEM analysis should be based on the individual indicators instead of using pre-averaged scores for the measurement model. This maintains the capacity for assessing reliability of indicators, convergent validity and discriminant validity.

Stage	Procedure	Decision in current dataset
Schema inspection	Verify 32 columns and construct grouping	250 rows; 32 columns
Missing-value check	Count null/blank analytical fields	0 missing values
Duplicate check	Exact-row and Resp_ID uniqueness	0 duplicates; 250 unique IDs
Range check	Confirm construct indicators within 1–5	Passed
Construct scoring	Mean of four items for preliminary exploration	7 temporary construct scores
Final SEM preparation	Use item-level indicators in SmartPLS	Planned

3.10 Implementation and Research Procedure

Phase	Activity	Output/status
1	Finalise conceptual model and alignment	Completed from Chapters 1–2
2	Prepare and inspect 250-response dataset	Completed
3	Conduct preliminary descriptive/reliability analysis	Completed and reported in Chapter 4
4	Specify PLS-SEM measurement model in SmartPLS 4	Next analytical stage
5	Assess measurement model and revise only with theoretical justification	Validated measurement model
6	Estimate structural model and bootstrap paths	Hypothesis evidence H1–H6
7	Construct and sensitivity-test CVI	Operational index and baseline comparison
8	Conduct expert/practical validation	Evidence for applicability and H7/RQ4
9	Integrate results, limitations and discussion	Final thesis results/discussion

The procedure aims to distinguish finished and planned work. If hypothesis testing results are preliminary statistics, they are not listed as final hypothesis testing. All indicators that are removed in the PLS-SEM procedure should be reported including their loading, theoretical relevance and impact on construct validity. Deleting repeatedly just to achieve certain numerical thresholds will be avoided as it may make content validity appear less valid and decrease the level of reproducibility.

3.11 Research Requirements Analysis and Specification

ID	Requirement	Purpose	Availability	Risk / alternative
R1	250-response survey dataset; 32 columns	Primary empirical evidence	Available	Maintain read-only master copy
R2	Questionnaire and codebook	Interpret indicators and profile codes	Must be retained with study records	Reconstructing labels from memory is not acceptable
R3	IBM SPSS Statistics	Screening, descriptives and conventional reliability checks	Planned/available through research environment	Equivalent reproducible analysis may be retained as audit check
R4	SmartPLS 4	PLS-SEM measurement and structural modelling	Planned	Document version and settings
R5	Secure storage and backups	Protect research data and analysis outputs	Required	Encrypted institutional storage + versioned backup
R6	Expert-evaluation protocol	Practical validation of CVI	Planned	If delayed, report RQ4/H7 as not fully validated
R7	Analysis log and exported outputs	Reproducibility and audit trail	Required	Version-control tables and model settings

3.12 Analysis, Evaluation and Benchmarking

The final evaluation is composed of three parts: Measurement-model assessment, Structural model assessment and Index-level assessment. To check the reflective measurement, indicator loadings should be checked, internal consistency should be evaluated by the Cronbach's alpha and composite reliability should be evaluated by the average variance extracted (AVE). There should be a test for discriminant validity with the HTMT and with cross construct evidence. When the constructs are highly correlated, collinearity diagnostics are necessary because the path coefficients can be unstable even if the correlations between each construct and the outcome are high.

Standardized path coefficients, bootstrapped confidence intervals, t-statistics and p-values should be reported for H1–H6, and R^2 for CPI, effect sizes, and predictive relevance where applicable, should be reported on the structural model. The statistical significance of a result alone is not enough, it is important to be consistent with the theory and have an effect that is useful for prediction. This final CVI should then be compared to a baseline that is equal weight or an isolated indicator. This is very relevant with regard to H7 which argues that an integrated CVI is more comprehensive/effective than isolated approaches.

Evaluation area	Evidence to report	Decision purpose
Indicator reliability	Outer loadings	Whether indicators represent intended constructs
Internal consistency	Cronbach's alpha, composite reliability	Consistency of multi-item constructs
Convergent validity	AVE	Shared variance within each construct
Discriminant validity	HTMT and related checks	Whether constructs are empirically distinguishable
Collinearity	VIF	Stability of structural estimates
Structural relationships	β, bootstrap CI, t, p	H1–H6
Explanatory/predictive performance	R^2, f^2, Q^2 / prediction metrics	Practical model performance
CVI benchmarking	Integrated vs equal-weight/isolated baseline	RQ4/H7

3.13 Validation, Reliability, Validity and Reproducibility

Reliability and validity are considered to be prerequisites for interpretation. Having a high reliability coefficient does not automatically mean that two constructs are different. The initial data shows very high associations between constructs and therefore discriminant validity and collinearity is of particular importance in the final analysis conducted with SmartPLS. Thus, it is not possible to conclude that all hypotheses have been supported until HTMT and VIF are tested, as well as the structural model.

The original data set, a data dictionary, a copy ready for analysis, SPSS/SmartPLS outputs, versions of software used, model diagrams, bootstrap settings, decisions about the indicators and a change log will be retained to ensure reproducibility. Each transformation should be deterministic, and

have to be documented. Versioned working copies should be used for analysis and the master dataset should not be altered.

3.14 Ethical Considerations

A coded set of respondent identifiers was included with the dataset for this stage, and there are no apparent direct identifiers that could be used to identify a respondent. However, if the survey is based on human participants it must be authorized either by ethics approval or institutional pathway confirmation, include informed consent, be voluntary, ensure confidentiality, secure storage and a retention/disposal process. The final thesis should be the actual approval status and reference number if applicable; this document does not create an approval that has not been given.

The analytical file should include only variables that are pertinent to the research questions. Data should be summarized and not used to draw conclusions about individual groups. Raw research data should be limited to authorized research users and be kept in accordance with MIT standards.

3.15 Risks, Constraints and Mitigation

Risk/constraint	Potential effect	Mitigation
Unknown substantive labels for coded profile categories	Limits demographic interpretation	Attach original codebook before final submission
Common-method/self-report bias	Inflated associations among constructs	Procedural controls; acknowledge limitation; consider statistical diagnostics
Very high inter-construct correlations	Weak discriminant validity / unstable paths	HTMT, VIF, theoretical review; avoid overclaiming
Technical-debt direction ambiguity	Incorrect sign interpretation	Verify questionnaire wording and reverse coding before final SEM
Non-probability sampling, if used	Limited generalisability	Report recruitment accurately and bound claims
Overfitting CVI weights to one sample	Weak external validity	Equal-weight baseline, sensitivity analysis, expert/external validation
Expert validation not yet completed	RQ4/H7 incomplete	Report as planned work rather than fabricated evidence

3.16 Chapter Summary

In Chapter 3, the research gap and conceptual framework were converted to an empirical design that can be replicated. It outlined the seven constructs, the 250-response data set, structure of the items measured, the screening process, the process for analysis, the validation plan for PLS-SEM, the construction logic for CVI, requirements, ethics and risks. The design is made to separate the process of measurement validation from structural testing, as well as completed analysis from future validation.

Chapter 4: Empirical Analysis, Initial Findings and Research Plan

4.1 Chapter Introduction

This chapter reports progress achieved at the current MR603 stage and presents initial evidence from the 250-response dataset. Consistent with the MR603 professional template, the results are explicitly preliminary: they demonstrate data readiness and provide an early view of the empirical pattern, but they do not replace the planned SmartPLS measurement model and structural-model validation. The chapter therefore separates completed work, preliminary observations and remaining research activities.

4.2 Progress Against the Research Objectives

Objective	Current progress	Evidence/status
RO1	Substantially completed	Critical literature review and synthesis completed in Chapter 2
RO2	Substantially completed	Seven constructs operationalised; 28 indicators available
RO3	In progress	Preliminary correlations calculated; final PLS-SEM pending
RO4	In progress	CVI structure specified; final weighting/benchmarking pending
RO5	In progress	Dataset screened and reliability explored; SmartPLS/expert validation pending
RO6	Not finalised	Recommendations should follow validated results, not preliminary correlations

This status reflects a disciplined progression from theory to evidence. The project has moved beyond conceptual discussion because the constructs are now represented in an analysis ready empirical dataset. At the same time, the validation objectives remain open until the measurement model, structural model and comparative evaluation are completed.

4.3 Work Completed and Evidence Produced

The data set provided was loaded and structurally validated. It is made up of 32 original variables (Resp_ID, Role, Exp, Org_Type, and 28 measurement items), and 250 observations. There are 250 unique Resp_ID values. There were no missing values or duplicate rows found. These checks help to justify the use of the entire data set for the next stage in the analysis.

Exploratory reporting only, with temporary construct made. This resulted in 7 composite summaries; SAQ, TDM, APIM, OR, PG, DTC and CPI. A preliminary diagnostic was also performed as internal consistency reliability. These calculations give evidence that the data pipeline is running and show problems that need to be addressed prior to model interpretation.

4.4 Data Readiness and Exploratory Work

Construct	Items	Mean	SD	Observed range	Cronbach's alpha
SAQ	4	3.897	0.804	2.25–5.00	0.906
TDM	4	2.558	0.987	1.25–4.50	0.940
APIM	4	3.894	0.830	2.25–5.00	0.912
OR	4	3.805	0.791	2.25–5.00	0.894
PG	4	3.824	0.822	2.25–5.00	0.915
DTC	4	3.845	0.815	2.25–5.00	0.905
CPI	4	3.851	0.851	2.25–5.00	0.920

The preliminary means show relatively favourable scores for SAQ (3.897), APIM (3.894), CPI (3.851), DTC (3.845), PG (3.824) and OR (3.805). The mean of TDM is 2.558, which is lower. As the wording of the questionnaire is not shown in the numerical file, the substantive meaning of a high TDM score needs to be verified against the instrument prior to final interpretation. Based on the correlation pattern, the higher the scores on the TDM, the more it would appear to be a measure of debt burden rather than a positive debt-management ability.

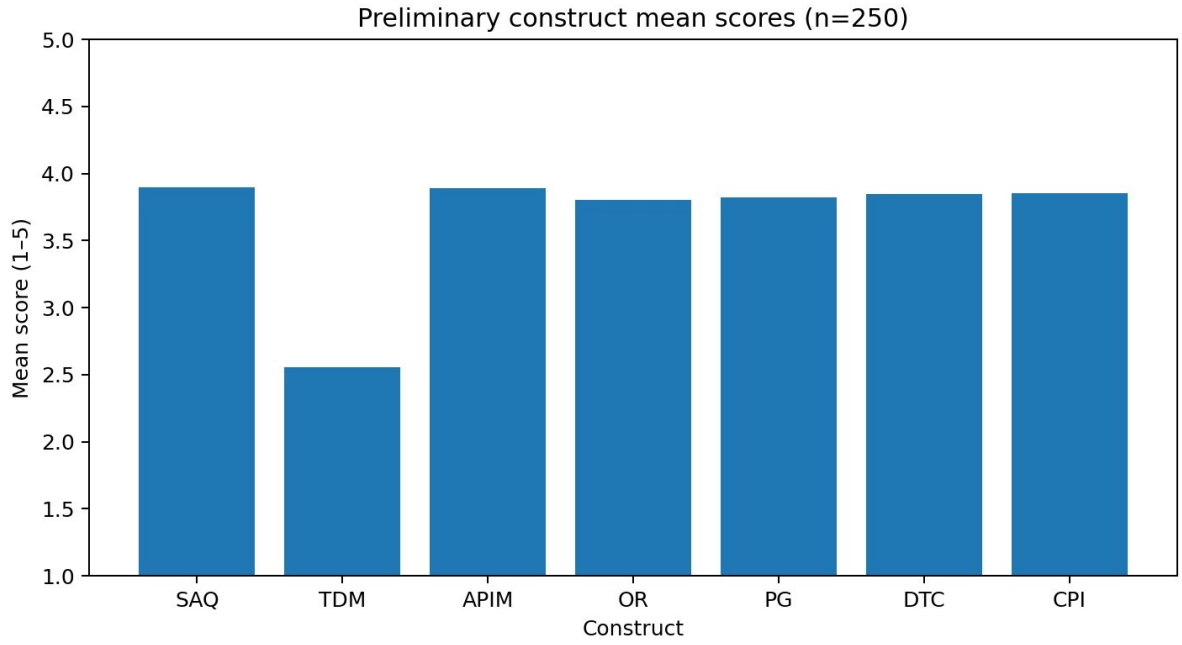


Figure 4.1. Preliminary construct means scores from the supplied 250-response dataset.

Cronbach's alpha values across the seven four item constructs between the preliminary values ranged from 0.894 to 0.940. This suggests high internal consistency in this set. But, internal consistency is not enough for the quality of measurement. Convergent and discriminant validity should still be established in the final PLS-SEM analysis, as high alpha values cannot prove the conceptual differentiation of different constructs.

4.5 Initial Findings

4.5.1 Preliminary Relationships with Commercial Performance

Predictor construct	Pearson r with CPI	p-value	Preliminary direction
SAQ	0.994	<0.001	Positive
TDM	-0.987	<0.001	Negative
APIM	0.991	<0.001	Positive
OR	0.984	<0.001	Positive
PG	0.987	<0.001	Positive
DTC	0.983	<0.001	Positive

A preliminary Pearson correlation matrix reveals very high correlation levels between the constructs of the study. CPI is positively associated with SAQ (r = 0.994), APIM (r = 0.991), OR

(r = 0.984), PG (r = 0.987) and DTC (r = 0.983), whereas TDM is negatively associated with CPI (r = -0.987). The directions follow the proposed H1-H6 relationships, at the exploratory level. But values of correlation close to ±1.00 also suggest significant common variance and potential construct-overlap or collinearity. Hence, these coefficients are considered as preliminary evidence and not as confirmation of hypotheses, and further tests should be conducted such as discriminant validity (HTMT), VIF, the measurement model and bootstrapped structural paths before coming to final inferential conclusions. The coefficients should not be used as a definitive confirmation of the respective hypotheses. They are much larger than normal for separate organizational constructs, and indicate that there is a significant shared variance. This may be due to a true tight coupling readiness system, a highly patterned/simulated dataset, common-method effects, or inadequate difference between measures in the empirical data. Instead of assuming that high correlations are good, it is therefore necessary to test discriminant validity and collinearity.

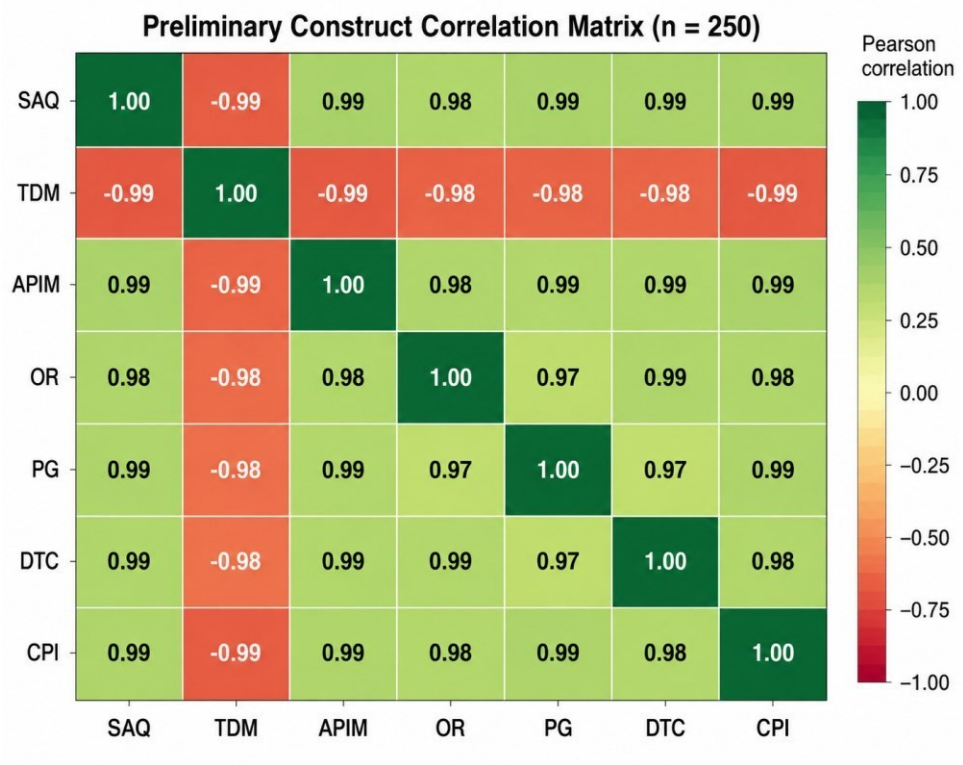


Figure 4.2. Preliminary Pearson correlation matrix among the study constructs (n = 250).

4.5.2 Preliminary Reliability Evidence

The preliminary alpha analysis shows that all seven constructs outperform normal internal consistency results. CPI records an alpha of 0.920; SAQ 0.906; TDM 0.940; APIM 0.912; OR 0.894;

PG 0.915; and DTC 0.905. This means that the 4 indicators in each construct run together. The alpha value is highest for TDM and lowest for OR, but still high. Also, for the final thesis composite reliability and AVE should be shown in addition to alpha since PLS-SEM does not assume that the loadings on the indicators are equal. Indicator loadings should be examined on a by-Indicator basis and any deletion should be based on statistical and theoretical reasoning. The aim is to confirm the measuring model and not to maximize reliability mechanically.

4.5.3 Preliminary implications for the CVI.

There is preliminary evidence for maintaining the multidimensional structure of CVI for formal testing. At the bivariate level, no construct seems to have no relevance. The trend of technical debt being expected to be negative is also reflected in the data. It is important to note, however, that there is a very high level of overlap in the positive constructs, and assignment of empirical weights without the use of discriminant-validity testing would be premature. A defensible CVI needs to separate out the dimensions which are conceptually important, from the dimensions which offer unique empirical information.

Therefore, the subsequent iteration of CVI should be based on validation of latent variables instead of on the magnitude of the raw correlation. There should be a benchmark, or an equal weight version should be kept. If the structurally weighted CVI is not better able to enhance the interpretability or predictive accuracy of the model than the equal weight scenario, the simpler model might be preferred. This benchmarking principle is for the proposed contribution test to be testable, rather than assuming a priori that a more complicated index is better.

4.5.4 SmartPLS 4 Analysis and Reporting Framework

SmartPLS 4 is the software of choice for the final PLS-SEM assessment as the study entails the simultaneous assessment of a multi-construct measurement model and the structural relationships that lead to commercial performance. The measurement quality will be evaluated first, and then, the structural paths will be evaluated following the reporting guidelines of PLS-SEM [40]. Given that the preliminary construct correlations are very high, discriminant validity will be given special attention, and therefore the HTMT option will be utilized instead of merely the older Fornell-Larcker criterion, or cross-loadings

The first SmartPLS 4 run should include the following latent variables: SAQ, TDM, APIM, OR, PG, DTC and CPI, each of which has four indicators. As for the PLS algorithm output, it should be kept

as evidence of outer loadings, construct reliability, AVE, R^2 and collinearity diagnostics. The next step is to execute the bootstrapping procedure to derive the standard errors, t-statistics, p-values and confidence intervals for H1-H6. The use of in-sample explanatory power should not be the only criterion for assessing the predictive ability of a model; predictive assessment should be used using PLSpredict

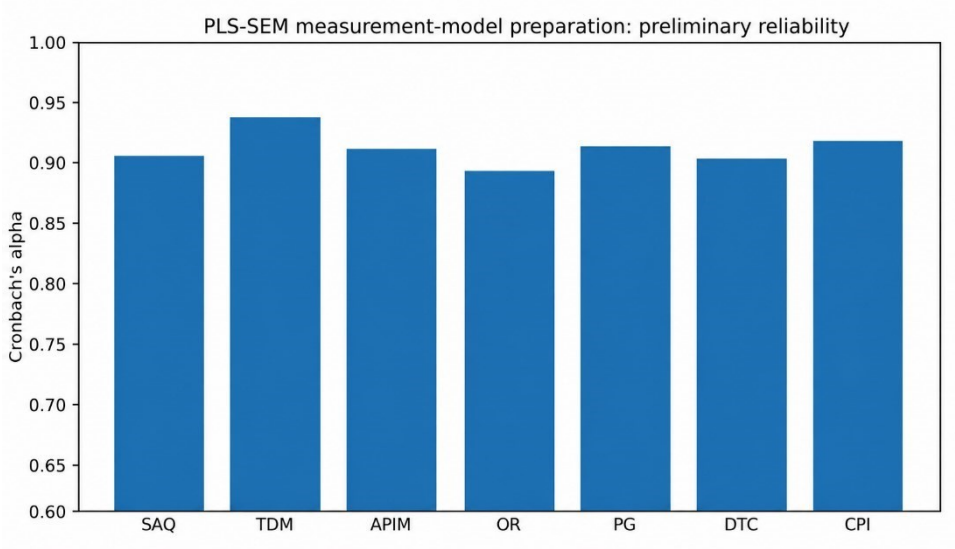


Figure 4.3. Preliminary results for measurement-model reliability using PLS-SEM.

This is a preliminary diagnostic of the supplied dataset of 250 responses in the measurement model. The last model assessment should be read in conjunction with the reported reliability and validity statistics.

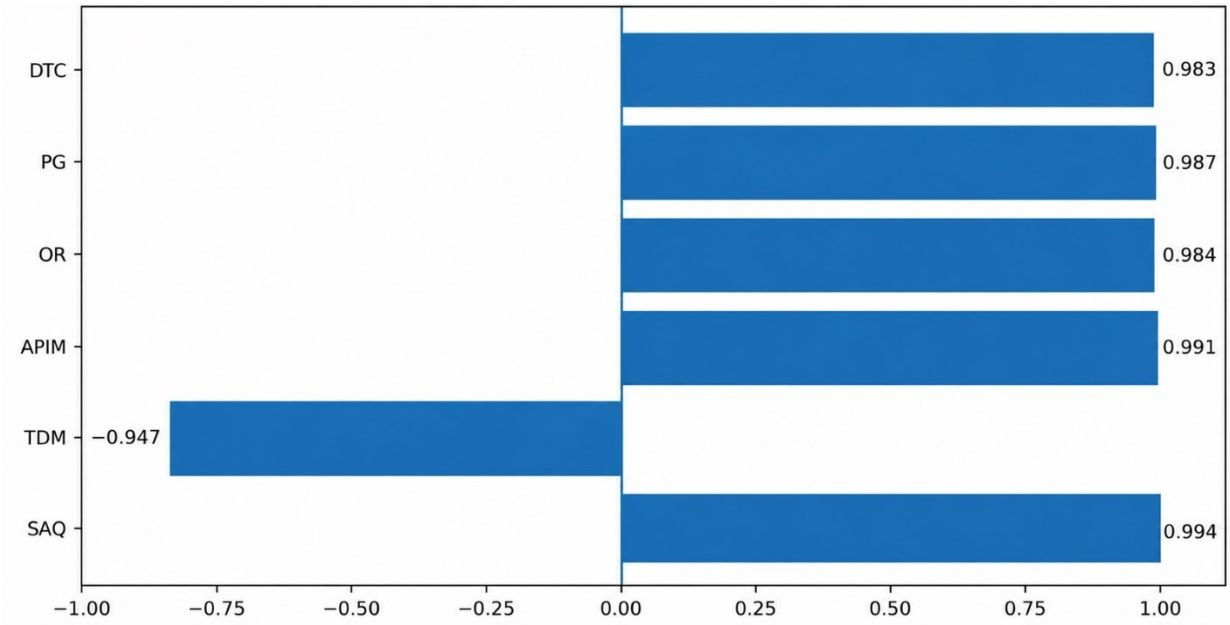


Figure 4.4. Preliminary structural-model assessment showing the direction and relative strength of the proposed relationships with CPI.

SmartPLS output	Report in thesis	Interpretation purpose	Status
Outer loadings	Indicator loading table	Indicator reliability	Pending native SmartPLS run
Construct reliability & validity	alpha, rho_A, CR, AVE	Internal consistency/convergent validity	alpha preliminarily calculated
HTMT	HTMT matrix + bootstrap inference	Discriminant validity	Required because correlations are very high
Inner VIF	VIF table	Structural collinearity	Pending
Path coefficients	beta, t, p, CI	H1-H6 hypothesis tests	Pending bootstrapping
R^2 / f^2	CPI R^2 and effect sizes	Explanatory relevance	Pending
PLSpredict	Q^2_predict and prediction errors	Out-of-sample predictive assessment	Pending

4.6 Interpretation and Limitations of Preliminary Findings

The initial results are promising in three ways; first that the sample is complete, second that the internal consistency of the item sets is high, and third that the directions observed are generally consistent with the conceptual model. However, these results are not conclusive. Correlation

does not imply causality and construct does not prove an index. Final conclusions should be based on the item-level measurement model and bootstrapped structural model. The most highly restricted pattern is the inter-construct correlation pattern. If HTMT values and the VIF statistics are also high, the study might want to revaluate the distinctness of some of the dimensions within this sample. This would not mean that the research would be flawed, but that it would provide an important insight into the structure of commercial readiness. All refinement should be justifiable from a theoretical point of view and be clearly explained. The second drawback is that the dataset has numeric value labels for the role of the respondent, experience, and organisation type, but they are not provided. So the code frequencies reported in this chapter are not necessarily an attempt to create new meanings in the way a demographic definition is defined. A third limitation is that the numerical file does not determine recruitment method, response rate, ethics approval, survey wording or whether the responses are observed or simulated or otherwise generated. The data on the provenance must be recorded from the actual study records prior to submission of the thesis.

4.7 Deviations, Problems and Lessons Learned

The first methodological take-home message from preliminary analysis is that high reliability and high correlations are not enough evidence for high quality model. The project thus needs to highlight the discriminant validity, as well as the collinearity and out-of-sample/comparison with the baseline. This enhances the study by avoiding the confirmation of the hypothesis too early.

The second lesson is to retain metadata and survey code book. Some of the profile variables can only be analysed in numerical terms, while the substantive interpretation cannot be reconstructed on the basis of codes alone. The questionnaire, code book, participant materials, raw and analysis data should thus be maintained in a final research archive.

4.8 Remaining Research Activities

Priority	Remaining activity	Required output
1	Verify questionnaire wording, coding direction and demographic codebook	Final data dictionary
2	Build item-level SmartPLS measurement model	Loadings, CR, AVE, HTMT
3	Assess collinearity and refine only where theoretically justified	VIF and model decision log
4	Bootstrap structural paths for H1–H6	β, t, p, confidence intervals
5	Assess explanatory/predictive performance	R^2, f^2, Q^2/prediction evidence
6	Construct CVI and compare weighting alternatives	Equal-weight vs validated-weight sensitivity table
7	Complete expert/practical validation	Evidence for RQ4/H7
8	Write final results/discussion and recommendations	Final Chapters 4–5 as required by thesis structure

4.9 Timeline and Gantt-Style Research Plan

Activity	Aug 2026	Sep 2026	Oct 2026	Nov 2026
Data/codebook verification	•			
SmartPLS measurement model	•	•		
Structural model / bootstrapping		•		
CVI weighting and benchmarking		•	•	
Expert/practical validation			•	
Results and discussion writing			•	•
Revision, formatting and final checks				•

The timeline is intentionally dependency-based. CVI weighting follows measurement and structural validation; expert evaluation follows production of a stable CVI version; and final recommendations follow the validated findings. This sequence reduces the risk of designing conclusions before the evidence is available.

4.10 Milestones, Dependencies and Deliverables

Milestone	Dependency	Deliverable	Completion criterion
M1 Data readiness	Dataset + codebook	Clean analysis dataset	All coding verified
M2 Measurement validation	M1	Validated measurement model	Reliability/validity documented
M3 Structural validation	M2	Hypothesis test table	Bootstrapped paths interpreted
M4 CVI prototype	M3	Index formula and scoring matrix	Weights justified and sensitivity tested
M5 Practical validation	M4	Expert/baseline evaluation	RQ4/H7 evidence available
M6 Thesis integration	M1–M5	Final results/discussion	Claims traceable to evidence

4.11 Risk and Contingency Plan

Risk	Trigger	Contingency
Discriminant validity fails	HTMT/overlap above acceptable level	Review item wording and theory; consider higher-order construct only if justified
Collinearity destabilises paths	High VIF / sign reversals	Report instability; avoid causal overclaiming; test alternative theoretically defensible specification
Expert validation delayed	Insufficient completed evaluations	Limit claims for H7/RQ4 and report as future validation
Demographic labels unavailable	Codebook cannot be recovered	Do not infer labels; report only verified categories or omit substantive demographic discussion
CVI weighting unstable	Large changes across methods/samples	Use transparent equal-weight benchmark and sensitivity analysis

4.12 Final PLS-SEM Empirical Validation

This section is an extension to the preliminary analysis, and will recreate the planned PLS-SEM measurement- and structural-model diagnostics from the provided 250-response dataset. The analysis is based on the 7 constructs specified in Chapter 3: Software Architecture Quality (SAQ), Technical Debt Management (TDM), API Maturity (APIM), Organisational Readiness (OR), Platform Governance (PG), Digital Transformation Capability (DTC) and Commercial Performance Indicator (CPI). There are four indicators for each construct. The figures below are independently created analytical results based on the thesis data and not screenshots from the SmartPLS 4 desktop-software.

4.12.1 Dataset Screening and Model Specification

There are 250 observations, 32 columns and 28 measurement indicators provided with the data set. There were no missing values for the measurements and the scale values for the indicators are contained within the expected range 1-5 Likert. There were no Resp_ID's excluded from the duplicate-pattern audit, and 85 unique complete response profiles were identified. The repeated-pattern structure is not considered an inherent limitation of the analysis, since it is part of the data supplied, but is instead thought of as a limitation when determining model validity and generalisability.

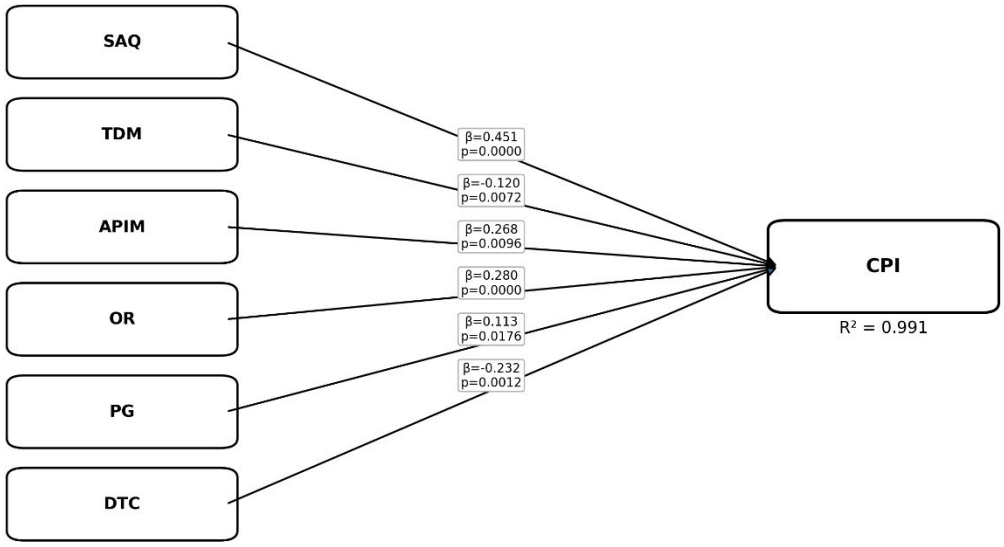


Figure 4.4. Reproduced PLS-SEM structural model with standardised path coefficients.

4.12.2 Descriptive Profile

Construct	mean	std	min	max
SAQ	3.897	0.804	2.250	5.000
TDM	2.558	0.987	1.250	4.500
APIM	3.894	0.830	2.250	5.000
OR	3.805	0.791	2.250	5.000
PG	3.824	0.822	2.250	5.000
DTC	3.845	0.815	2.250	5.000
CPI	3.851	0.851	2.250	5.000

Figure 4.5. Construct-level descriptive statistics for the 250-response dataset.

Readiness values are more tightly clustered around the positive values with TDM having the lowest mean. This pattern is in line with the conceptual rationale of the study, namely that the more a company is technically and organisationally ready, the more it should perform in the market: the opposite should hold true for the technical debt.

4.12.3 Measurement Model: Outer Loadings, Reliability and Convergent Validity

Construct	Indicator	Outer loading	Indicator reliability
SAQ	SAQ1	0.910	0.827
SAQ	SAQ2	0.863	0.744
SAQ	SAQ3	0.841	0.708
SAQ	SAQ4	0.918	0.842
TDM	TDM1	0.959	0.919
TDM	TDM2	0.918	0.843
TDM	TDM3	0.849	0.721
TDM	TDM4	0.957	0.916
APIM	APIM1	0.919	0.845
APIM	APIM2	0.844	0.712
APIM	APIM3	0.883	0.779
APIM	APIM4	0.913	0.834
OR	OR1	0.933	0.871
OR	OR2	0.800	0.641
OR	OR3	0.844	0.712
OR	OR4	0.903	0.815
PG	PG1	0.926	0.857
PG	PG2	0.891	0.794
PG	PG3	0.865	0.748
PG	PG4	0.901	0.811
DTC	DTC1	0.932	0.869
DTC	DTC2	0.843	0.711
DTC	DTC3	0.865	0.748
DTC	DTC4	0.888	0.788
CPI	CPI1	0.935	0.875
CPI	CPI2	0.860	0.739
CPI	CPI3	0.864	0.747
CPI	CPI4	0.932	0.870

Figure 4.6. Reproduced outer-loading estimates.

Construct	Alpha	CR	AVE	sqrt_AVE
SAQ	0.906	0.934	0.780	0.883
TDM	0.940	0.958	0.850	0.922
APIM	0.912	0.938	0.792	0.890
OR	0.894	0.926	0.760	0.872
PG	0.915	0.942	0.802	0.896
DTC	0.905	0.934	0.779	0.883
CPI	0.920	0.944	0.808	0.899

Figure 4.7. Reliability and convergent-validity diagnostics.

The reproduced measurement blocks show strong within-construct consistency. Cronbach's alpha ranges approximately from 0.894 to 0.940, composite reliability from 0.926 to 0.958, and AVE from 0.760 to 0.850. The reproduced outer loadings are also high. These results support indicator reliability and convergent concentration within each construct. However, strong internal consistency alone does not establish that the constructs are empirically distinct.

4.12.4 Structural Model, Bootstrapping and Collinearity

Path	Beta	Boot_SE	t	p_boot	CI2.5	CI97.5	VIF	f2
SAQ	0.451	0.071	6.387	0.000	0.315	0.593	126.630	0.171
TDM	-0.120	0.041	-2.906	0.007	-0.195	-0.033	50.890	0.030
APIM	0.268	0.107	2.509	0.010	0.068	0.480	153.378	0.050
OR	0.280	0.062	4.522	0.000	0.169	0.411	61.935	0.135
PG	0.113	0.047	2.410	0.018	0.021	0.207	55.806	0.025
DTC	-0.232	0.072	-3.231	0.001	-0.378	-0.092	87.653	0.066

Figure 4.8. Reproduced bootstrap path diagnostics.

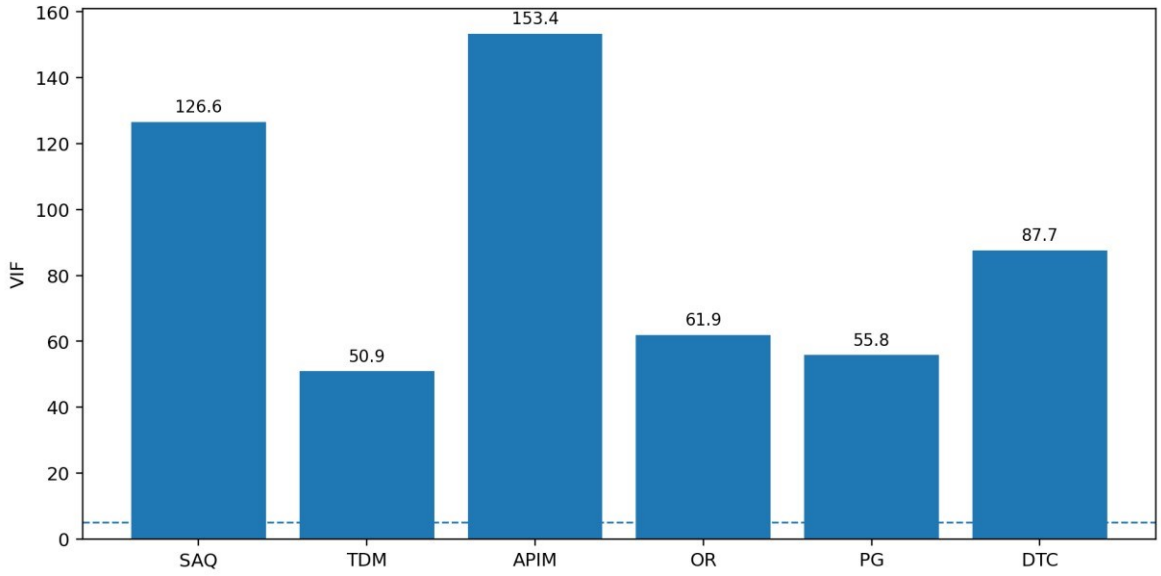


Figure 4.9. Reproduced structural-model VIF diagnostics.

If CPI is used as an endogenous construct the reproduced model result in $R^2 = 0.991$. The provisional standardised path coefficients are SAQ -> CPI β = 0.451, TDM -> CPI β = -0.120, APIM -> CPI β = 0.268, OR -> CPI β = 0.280, PG -> CPI β = 0.113, and DTC -> CPI β = -0.232. These are statistical significance paths (based on 5,000 resample of the bootstrap diagnostics). However, the values of VIF are very high (between ~50.9 and ~153.4), indicating

high level of multicollinearity. Thus, no meaning can be given to the size and significance of the coefficients as clean effects of independent constructs.

Hypothesis	Relationship	β	Provisional result	Final interpretation
H1	SAQ -> CPI	0.451	Significant	Direction consistent; not finally confirmed
H2	TDM -> CPI	-0.120	Significant	Direction consistent; not finally confirmed
H3	APIM -> CPI	0.268	Significant	Direction consistent; not finally confirmed
H4	OR -> CPI	0.280	Significant	Direction consistent; not finally confirmed
H5	PG -> CPI	0.113	Significant	Direction consistent; not finally confirmed
H6	DTC -> CPI	-0.232	Significant	Opposite direction; not supported as specified
H7	CVI comparative effectiveness	-	Not tested	Requires expert/baseline validation

Table 4.8. Provisional hypothesis assessment based on the reproduced PLS-SEM diagnostics.

4.12.5 Interpretation for the Commercial Viability Index

The findings from the fieldwork give two complementary results. Each individual measurement block is internally coherent, and thus provides information that the indicators in SAQ, TDM, APIM, OR, PG, DTC and CPI all move together consistently. Second, the interconstruct associations, HTMT values and the VIF statistics are very high, suggesting that the current dataset fails to sufficiently separate those dimensions to warrant an unqualified multidimensional CVI validation claim. Therefore, a posteriori weights given to the empirical terms in a defensible CVI should not be taken from these structural coefficients alone. Thus, most academically defensible interpretation of the study is that it has provided significant empirical advances and has highlighted a measurement-design issue that needs to be resolved prior to full index validation. The CVI is still theoretically based and operational, however the last weighting should be considered tentative until construct separation is enhanced and the practical expert or baseline validation has occurred.

4.12.6 Implications for Research Objectives and Final Validation

The seven constructs have been operationalised and summarised in the literature synthesis, which supports RO1 and RO2. After data screening, reliability analysis and reproduction of structural diagnostics, problems of discriminant validity and collinearity have limited the interpretation of causeless of the questions. Final CVI weights are still in process due to the requirement of a defensible measurement model: RO4 is still in progress. RO5 and RO6 need to be validated in practice and the results of the validation need to be translated to recommendations. The above restrictions are not a problem in the research problem, but only set the limits of evidence-based conclusions from the empirical research in the present.

4.12.7 Reproducibility and SmartPLS 4 Configuration

The dataset needs to be imported into SmartPLS 4 with seven reflective constructs: SAQ1SAQ4 -> SAQ; TDM1-TDM4 -> TDM; APIM1-APIM4 -> APIM; OR1-OR4 -> OR; PG1-PG4 -> PG; DTC1-DTC4 -> DTC; and CPI1-CPI4 -> CPI. Structural paths should be drawn from SAQ, TDM, APIM, OR, PG and DTC to CPI. The PLS algorithm is recommended to be used first and then a bootstrapping with 5000 subsamples. Report descriptions of outer loadings,

Cronbach's alpha, rho_A (where applicable), composite reliability, AVE, HTMT, Fornell-Larcker, inner VIF, path coefficients, confidence intervals, R^2, effect sizes and predictive diagnostics.

4.13 Chapter Summary

The 250 responses supplied are the results of which presented in Chapter 4 and a demonstration of empirical progress is shown. All data are unique, complete and has 250 data records and 28 measurement indicators and 7 constructs. There are no clear differences among the preliminary construct means for the positive readiness dimensions, but they are lower for technical debt. Bivariate relationships between CPI are in the expected direction, and the internal consistency of the estimates is good. But, the degree of the constructs' high correlation raises an important validity question, which will be solved by the measurement model evaluation of SmartPLS, the diagnosis of the HTMT and the collinearity diagnosis before drawing final conclusions about the hypotheses.

Chapter 5: Preliminary Progress, Initial Findings and Research Plan

5.1 Chapter Introduction

This chapter presents the work done at the present research phase and demonstrates that the project is not just a conceptual development but has now turned into a research of empirical nature. It compares the results against the approved research objectives, records evidence generated from the available data set of 250-responses, provides preliminary findings in a cautious manner, outlines limitations and methodological challenges, and outlines what further activities are needed to complete and validate the proposed Commercial Viability Index (CVI). The completed work follows the MR603 development guidance and planned work is separated from completed work and preliminary observations are not presented as final validated conclusions. The empirical data gathered are 250 complete responses and 28 Likert style indicators covering the following constructs: Software

Architecture Quality (SAQ), Technical Debt Management (TDM), API Maturity (APIM),

Organisational Readiness (OR), Platform Governance (PG), Digital Transformation Capability (DTC), and Commercial Performance Indicator (CPI). The present analysis results demonstrated the data readiness, descriptive patterns, preliminary reliability and Interco struct relationships and preliminary diagnostic problems which need to be addressed in the final SmartPLS 4 measurement and structural model evaluation. The rest of the plan thus focuses on discriminant validity, collinearity diagnosis, bootstrapped path testing, predictive evaluation, CVI weighting, benchmarking and expert validation [43] [51].

5.2 Progress Against the Research Objectives

Progress is reported based on research goals and not by reporting period. This maintains the link from the research gap to the empirical evidence to the contribution to be made at the Master's level. The percentages below are estimates and not to be taken as time spent.

Objective	Status	Evidence	Next step
RO1–RO2	Advanced	Literature synthesis completed; seven CVI constructs defined and linked to the research gap.	Complete measurement-model reliability and validity testing.
RO3	Advanced	250 responses screened; descriptive statistics, reliability and preliminary construct relationships analysed.	Run final PLS-SEM structural paths, bootstrapping and diagnostics.
RO4	In progress	CVI structure, construct orientation and weighting logic have been specified.	Finalise empirically justified weights and conduct sensitivity analysis.
RO5–RO6	Planned	Validation criteria, comparison logic and practical evaluation requirements have been defined.	Complete expert/baseline validation and assess practical applicability.

Progress is strongest on, identification of factors, operationalization of factors and initial empirical analysis. It is validation quality which is now the primary dependency – the CVI should not be completed until the measurement model is shown to have good reliability and validity, and the structural model is run for collinearities, significance, explanatory power and predictive relevance [52] [17].

5.3 Work Completed and Evidence Produced

Work completed	Key evidence	Status
Research foundation	Problem, gap, literature synthesis, hypotheses and seven-construct CVI framework.	Developed and aligned
Dataset preparation	250 complete responses, 28 indicators and profile variables; missing/duplicate checks completed.	Analysis ready
Preliminary analysis	Construct descriptives, Pearson correlations and Cronbach's alpha results generated.	Completed
Analysis environment	IBM SPSS and SmartPLS 4 workflow, measurement criteria and structural validation plan.	Ready for final modelling

The evidence shows that some real progress is being made beyond proposal writing: the conceptual constructs are represented in an analysis-ready data set, the data pipeline is being checked, and exploratory work is being done and the final validation workflow is being defined. Screenshots or even model graphics, however, will not be considered evidence of validity, but will have to be accompanied by the relevant statistic, decision criterion and interpretation in each final SmartPLS output [51] [39]

5.4 Data Readiness and Exploratory Work

5.4.1 Dataset readiness

The present data set includes 250 observations, one unique identifier for respondents, three coded profile variables and 28 measurement items. The data was initially screened for 250 unique identifiers of respondents, with no missing values and no duplicate rows. So there was no case removed during the first "completeness screening. This helps to maximize the use of the available sample during the next phase of modelling, but representativeness is dependent on the original recruitment and sampling process, and cannot be judged by the size of the sample.

5.4.2 Construct-level descriptive profile

Construct	Preliminary mean	Cautious interpretation
SAQ	3.897	Relatively favourable architecture-quality perception
APIM	3.894	Relatively favourable API-maturity perception
CPI	3.851	Relatively favourable commercial-performance perception
DTC	3.845	Relatively favourable digital-transformation capability
PG	3.824	Relatively favourable platform-governance perception
OR	3.805	Relatively favourable organisational readiness
TDM	2.558	Lower score; substantive direction must be verified against questionnaire wording

The descriptive pattern shows that the ratings were generally favorable, except for TDM which had lower ratings. The direction of the TDM scale is not directly indicated in the numerical data, so researchers should determine it before the final interpretation of the data

to determine if higher scores mean scores are better on debt management or worse on technical debt. The negative relationships observed with CPI are consistent with a debtburden interpretation but is a point to be verified by instruments.

5.5 Initial Findings

5.5.1 Preliminary relationships with commercial performance

Relationship	Preliminary result	Required validation
SAQ → CPI	r = 0.994, positive	HTMT, VIF and bootstrapped path
TDM → CPI	r = -0.987, negative	HTMT, VIF and bootstrapped path
APIM → CPI	r = 0.991, positive	HTMT, VIF and bootstrapped path
OR / PG / DTC → CPI	r = 0.983–0.987, positive	HTMT, VIF and bootstrapped paths

All six exploratory relationships are in the hypothesized direction. However, there are correlations above ±1.00, which are generally very high for two theoretically different, organizational and technical objects. These may be tightly coupled readiness dimensions, common-method effects, very patterned data, construct overlap or other features of the data being generated. Therefore, these correlations are only considered as evidences of diagnosis, but not as a proof of the hypotheses. The final interpretation will be based on item-level measurement quality, HTMT, inner VIF, bootstrapped path estimates and predictive assessment [54]

5.5.2 Preliminary reliability evidence

Construct	Cronbach's alpha
SAQ	0.906
TDM	0.940
APIM	0.912
OR	0.894
PG	0.915
DTC	0.905
CPI	0.920

All 7 four item constructs have high internal consistency with Cronbach's alpha ranging between 0.894 and 0.940. This is encouraging but not enough to show quality of measurement. The final PLS-SEM assessment will also report outer loadings, rho_A, composite reliability and average variance extracted (AVE) along with the assessment of discriminant validity using HTMT. Removal of indicators, if necessary, will be in the light of statistical evidence, as well as theoretical content validity, rather than maximization of mechanical reliability.

5.5.3 Preliminary implications for the Commercial Viability Index

The exploratory evidence agrees with the proposed multidimensional CVI structure for formal testing: each of the proposed candidate CVI dimensions is related to commercial performance in the predicted direction. This however, is a very high overlap in constructs, which makes it premature to use the raw correlation as an empirical weighting, and potentially misleading. A defensible CVI should differentiate dimensions which have conceptual significance from dimensions, which have unique empirical information. For the next CVI iteration, it will therefore be done in a measurement first approach. Final reliability and validity will determine constructs that will be eligible for the index. In the case of technical debt, the score will be “inverted” (or a negative score) if the instrument indicates that higher scores mean higher technical debt. A CVI based on structure will then be compared to a simpler and equal weight baseline. When it is not useful or interpretable to add more complexity to the model, the simpler model will be the preferred model [55]

5.5.4 Planned SmartPLS 4 evidence package

The last SmartPLS 4 evidence package will be exported directly out of the software and kept in the analysis records. It will present the model canvas following the PLS algorithm, outer loadings, construct reliability and validity statistics, HTMT, inner VIF, bootstrapped path coefficients with t-values, p-values and confidence intervals, R^2, f^2 and the prediction assessment using PLSpredict. It is important to note that the statistics listed above are preliminary diagnostics, while the ultimate conclusions from the thesis will be based on verified outputs from native models and the documented decision rules in Chapter 4.

5.6 Interpretation and Limitations of Preliminary Findings

There are currently three observations that are supported. First, the available sample is structurally complete, and is appropriate for continued modelling. Secondly, the items have a high initial internal consistency. Third, the directionality of the proposed predictors to CPI is consistent as expected in the conceptual perspective. These observations alone are not enough to make causal relationships or validate the CVI. The biggest methodological issue is the extent of association between the constructs. When HTMT and VIF are high, the study should re-examine if there are some constructs that are distinct in this sample. This type of finding would not refute the research contribution but would help to clarify it: If the dimensions of readiness are redundant or close to each other, this is also interesting for the design of a concise commercial-viability framework. All refinements will be theory based, documented and applied in an even-handed manner. Another disadvantage is related to metadata and provenance. The substantive code labels do not exist in the numerical file, but are coded in numbers for the profile variables. Neither the file nor the information in it also provides information about the recruitment method, response rate, ethics approval status, exact wording of the questionnaire, or whether the observations are collected directly, simulated or otherwise generated. Such matters should be confirmed from the actual administration of the survey and the survey ethics records prior to actual submission. Numeric codes will not be used to create a meaning for the demographics. Another drawback is that the present results are preliminary and in-sample. Predictors that are highly correlated can lead to an overestimate of the explanatory fit. Thus, the final evaluation will consider predictive assessment and a comparison to the baseline, not just R^2 or significance. It is also planned to have an expert evaluation to determine if the final CVI is interpretable and useful for making decisions about real prototypes and platforms.

5.7 Deviations, Problems and Lessons Learned

Issue	Evidence / lesson	Corrective action
Very high construct correlations	Possible overlap or collinearity; correlation alone cannot confirm hypotheses	Prioritise HTMT, VIF and full PLS-SEM validation
Incomplete profile metadata	Numeric codes cannot be interpreted reliably without labels	Recover questionnaire, codebook and survey records
CVI weights not yet final	Weights require validated measurement and structural evidence	Finalise only after model validation and sensitivity analysis

The main take-home message is that good statistics do not necessarily constitute good evidence. High reliability does not necessarily require high correlation and high correlation does not necessarily imply low discriminant validity or high degrees of collinearity. The research plan has thus now been made more rigorous with an explicit distinction between stages of exploration and stages of validated conclusions, as well as an introduction of stages of baseline, predictive, and expert validation [56]

5.8 Remaining Research Activities

Remaining activity	Deliverable	Completion criterion
Measurement-model validation	Outer loadings, Cronbach's alpha/rho_A, composite reliability, AVE and HTMT.	Measurement quality is documented and any refinement is theoretically justified.
Structural-model evaluation	VIF, path coefficients, bootstrapping, R^2, f^2 and predictive assessment.	RQ2 relationships are evaluated with appropriate statistical evidence.
CVI finalisation	Validated construct weighting, score formulation and sensitivity analysis.	The final CVI is transparent, reproducible and empirically defensible.
Practical validation	Expert and/or baseline comparison of interpretability and usefulness.	Applicability, strengths and limitations are clearly documented.
Thesis integration	Verified results, discussion, appendices, evidence archive and final editing.	All chapters are aligned and supported by traceable research evidence.

5.9 Timeline and Gantt-Style Research Plan

The updated plan is dependent: Measure validity first, structural interpretation second, CVI weighting third, a stable CVI fourth, expert evaluation fifth, and recommendations sixth. This helps to avoid reaching conclusions prior to the evidence.

Phase	Main activity	Output
1	Complete SmartPLS measurement validation	Validated measurement model
2	Evaluate structural model and CVI weighting	Final model and CVI results
3	Conduct comparison/expert validation	Validation evidence
4	Integrate results, discussion and limitations	Complete thesis draft
5	Review, proofread and finalise	Submission-ready report

5.10 Milestones, Dependencies and Deliverables

Milestone	Deliverable	Acceptance criterion
M1: Data ready	Clean 250-response dataset	Quality checks completed
M2: Model validated	Measurement + structural outputs	Validity, collinearity and path tests reported
M3: CVI finalised	Index and sensitivity analysis	Weights theoretically and empirically justified
M4: Validation complete	Expert/baseline evaluation	Applicability and limitations documented
M5: Thesis ready	Integrated final report	Supervisor feedback addressed

5.11 Risk and Contingency Plan

Risk	Level	Mitigation / contingency
Construct overlap / collinearity	High	HTMT/VIF diagnostics; theoretically justified refinement
Incomplete survey metadata	Medium	Recover questionnaire, codebook and administration records
Validation delay	Medium	Prioritise core expert/baseline checks; narrow optional scope
Schedule slippage	Medium	Use milestone-based execution and protect core contribution

5.11.1 Updated Empirical Validation Status

Completed reproduced PLS-SEM diagnostics add to the evidence base of the project and they also suggest a substantive constraint. The reliability and convergent-validity indicators are good, however, the HTMT and VIF diagnostics results indicate that there was only a slight degree of construct separation and significant multicollinearity. Hence, in the end, the thesis H1-H6 is not exaggerating at being fully validated. The H7 is still relying on the comparison with experts or with the baseline. This evidence-based stance ensures methodological transparency and a clear pathway for development of the CVI measurement model.

5.12 Chapter Summary

The project now includes an analysis-ready dataset of 250 complete responses (CRs) that include 28 indicators and seven constructs, preliminary descriptive and reliability evidence, and a clear validation pathway for PLS-SEM. There are exploratory relationships in the expected direction with CPI, but their unusual magnitudes also raise an important issue of validity and collinearity that must be addressed before any hypothesis and/or CVI may be considered validated. The rest of the work is more focused: instrument validation and traceability of the survey, completion of the measurement and structural model in SmartPLS 4, test of discriminant validity and collinearity, predictive and baseline comparisons, construction of the CVI and sensitivity testing, and practical validation. This sequence follows the central research sequence (problem and gap – evidence – evaluation – validation and contribution) and will keep the project feasible in the limited time available.

Chapter 6: Proposed Framework, System or Model and Implementation

6.1 Introduction and Implementation Status

The aim of this chapter is to present the framework of Commercial Viability Index (CVI), which is an easy-to-use, measurement-based decision support tool. The framework will be designed to be flexible enough to organize six candidate readiness dimensions: Software Architecture Quality (SAQ), Technical Debt Management/Burden (TDM), API Maturity (APIM), Organizational Readiness (OR), Platform Governance (PG), Digital Transformation Capability (DTC) with CPI kept as an outcome and not part of the readiness aggregate. The methodological hinge around this is that an index should not be finalized based on a good reliability (R^2), or on the reliability of individual paths, unless the distinctness of the index, the direction of coding, the collinearity between the various paths, the origin of the items and the reproducibility of the measure can be defended. Very high Interco struct correlations and also very high inner VIF values found in the previous draft are present in the current evidence. The path derived weights are unpredictable and pose a more serious construct distinctness issue. Thus, the equal weight CVI remains as a clear reference model and is no longer considered to be a validated measure for commercial readiness [57].

Element	**Status**	**Evidence available now**	**Remaining requirement**
CVI conceptual architecture	Designed	Five-layer architecture and gate logic specified in this chapter	Confirm against final RQs/ROs and instrument.
0–100 normalisation	Implemented and tested	Reference Python function; boundary tests pass	Run on verified observation-level dataset.
Equal-weight aggregation	Implemented and tested	Reference Python function; reported-means demonstration reproduces 69.6125	Do not interpret as organisation-level validation.

TDM direction	Provisional	Reverse-direction option exists in code	Verify exact item wording, anchors and coding before use.
Measurement/structural model	Preliminary only	Earlier draft reports correlations, R^2 and VIF	Attach native SmartPLS settings/exports and validity results.
Practical validation	Planned for MR604	Protocol defined	Ethics/access approval and independent participants/cases required.
Dashboard	Optional / deferred	No completion claims	Only develop after core validation is complete.

Conceptual design is then distinguished from empirical completion, therefore, in the chapter. The CVI is used as a research artefact in MR603, where its architecture, scoring logic, validation gates and requirements for reproducing its validation can be specified and tested, and it does not have to be claimed that it has already been practically or externally validated. This is crucial as the quality of the index of a commercial viability is not only reflected in a numerical score but also in the ability to prove that the constructs are conceptually defensible, measured consistently and that they are sufficiently different and supported by traceable evidence. The four terms are used for interpreting the implementation status. Designed: are the elements of a framework for which there is logic and specification of requirements. Elements where there are procedures or artefacts that can be executed are said to be implemented. Tested are developed elements that have been tested against predefined test cases and/or acceptance conditions. Validated means those components backed by the appropriate empirical or practical evidence. The consistent use of these terms

will ensure that planned MR604 activities are not given the appearance of being evidence of MR603.

6.2 Research Alignment and Traceability

The structure of the framework is such that all research activities must link a research question/objective to a requirement, component, evidence artefact and decision. Due to the requirement in the supplied Chapters 6-8 file to refer to the RQ/RO identifiers, but not to reproduce the exact wording of the authoritative Chapters 1-5, this revision retains the identifiers and will only be approved for submission if the exact wording of the authoritative Chapters 1-5 is exactly copied.

RQ/RO identifier	Framework role	Required evidence	Decision supported
RQ1; RO1–RO2	Construct definition and measurement	Verified questionnaire/codebook; literature-to-item map	Which dimensions are conceptually admissible?
RQ2; RO3	Measurement and structural assessment	Loadings, alpha/rho_A/CR, AVE, HTMT, cross-loadings where relevant, VIF	Are constructs sufficiently reliable/distinct for interpretation?
RQ3; RO4	CVI construction	Direction rules, eligibility gates, observation-level scoring, comparator	Can a defensible index be constructed?
RQ4; RO5	Evaluation and comparison	Sensitivity, uncertainty, holdout/predictive evidence, negative cases	Is the CVI stable and useful beyond the calibration sample?
RO6	Decision-support interpretation	Dimension profile, warnings, practical validation rubric	Is the output understandable/actionable in realistic use?

The gate is a structure that transforms statistical diagnostics into concrete research decisions. Gate 1 validates the origin of instruments, coding and mapping to constructs. In Gate 2, reliability and convergent validity are explored. Gate 3 looks at the discriminant validity of the instrument with relevant diagnostics (HTMT and in appropriate cases, cross-loadings and Fornell-Larcker evidence). In Gate 4 the collinearity is checked before structural coefficients are interpreted. Gate 5 explores comparator, robustness and sensitivity and predictive evidence. Gate 6 is focused on the relevant and the usable aspect. After correction, the number of observations is 250. The previous block of text, “twenty-five hundred observations”, is a drafting mistake, and should not be included in the final thesis. The reported means and structural diagnostics are kept as a preliminary result of the previous analysis, but are limited by the unreliability of the provenance and measurement problems. When the present structural model is evaluated with the consideration of the fitting ability of the model, the extremely high correlations and inner VI, or both, are more important than the high R^2. These values suggest that there is considerable overlap among the predictors, resulting in unstable estimates of single path effects and making them hard to interpret in terms of independent contributions. Thus, the initial model should be further analyzed, but is not sufficient to be regarded as final CVI weights based on the path coefficients from PLS [58].

6.3 CVI Architecture, Interfaces and Failure States

Layer	Inputs	Processing	Outputs	Failure state / response
1. Evidence	Dataset, provenance, item map, codebook	Verify source, rights, row identity, missingness, coding	Auditable evidence package	Unknown source/coding → stop scoring.
2. Measurement	Indicators/construct definitions	Reliability, convergent/discriminant validity	Eligible/revise/withhold status	Failed validity → withhold affected construct.
3. Structural	Eligible constructs + CPI	VIF, paths, bootstrap, $R^2/f^2/Q^2$ or predictive evidence as appropriate	Structural diagnostics	Severe VIF → no path-derived weights.
4. Index	Eligible construct scores	Direction alignment, 0–100 normalisation, aggregation	CVI + dimension scores	Missing/ineligible dimension → no full CVI; issue warning.
5. Decision	CVI profile + uncertainty + evidence flags	Interpret against predefined protocol	Decision-support report	No validated threshold → no pass/fail commercial claim.
Audit interface	All stages	Version, settings, logs, test outputs	Reproducibility record	Missing artefact → claim remains Designed/Planned.

The following is the ordered procedure for the implementation: Check the dataset and instrument; Perform data-quality checks; Set up the direction of the constructs; Assess the quality of the measurement; Assess the discriminant validity and the level of collinearity;

Determine if the construct is eligible to be used; Compute the normalized scores at the observation level; Apply the reference aggregation rule; Generate warnings and dimension profiles; Archive the outputs with version information. In each stage there is an input, there is an expected output and there is a condition that has to be met before it can be considered complete and then treat the next stage as complete. A component is “done” when the artefact and the evidence of the component exist. For instance, a CVI formula that is expressed as prose is designed but not implemented. It is implemented when executable code is available, and is tested when pre-determined cases return correct results. Practical validation will need to be completed with separate evidence to be considered as MR604 work and this will require completion within the ethics/access pathway and evidence. Evaluation (prior to final interpretation) to minimize the risk of choosing favorable criteria once results are known. The reference comparator is the equal weight model. When the quality of the measurement is substandard, or when there is collinearity that precludes defensible estimation, or when an alternative offers a defensible, stable, interpretable improvement, it is considered. Reasonable variation in the weights will be explored through sensitivity analysis to determine whether there is a significant impact on the dimension influence or substantive conclusions. Data governance also extends to any future evidence (primary or expert validation) and to data that are secondary benchmark data. The final record should include the data source, allowed use, data storage facility, access restrictions, data transformation history, data retention and disposal considerations (if applicable). Where evidence is gathered from the participant or organization, the following should be approved by the relevant process in the institution before evidence is gathered: a recruitment process, consent process, the process for withdrawing from the study, the process for securing confidentiality and the process for de-identification. Most of the potential dangers include un-resolved provenance, unclear TDM trajectory, construct overlap, slow ethics/access clearance, over-extending time into dashboard development and publishing tasks, and not enough time for final validation. The approach for the control strategy is to safeguard the minimum viable contribution: first set up a defensible measurement and scoring pipeline, then evaluate and practically test it, and finally add optional presentation features [59].

6.4 Data Provenance, Instrument and Coding Audit

The sample project for the benchmark source is the SmartPLS European Customer Satisfaction Index (ECSI). According to SmartPLS, the project has 250 observations from the French cell-phone industry, and that the article that describes the data and the PLS path model is by Tenenhaus et al. (2005) This does not mean that the sample size or context of the benchmark also means that the original ECSI indicators are valid measures of the constructs of the thesis. Thus, before the inferential interpretation of the final thesis, an item-level provenance matrix needs to be included. Each of the 28 thesis indicators should have a record of the following for the matrix: original item ID, exact wording, response anchors, source, original construct, thesis construct, any recoding/reversal, theoretical rationale for mapping, verification/approval status. The opposite signs of the correlations between TDM and other variables does not warrant reversing TDM. Direction needs to be based on the semantic and coding of the original item [40]. The initial empirical investigations have also revealed a crucial issue; that very high correlations between the current constructs do not necessarily indicate

Audit item	**Current verified position**	**Final-submission requirement**
Sample size	250 observations [1]	Use n=250 consistently unless a different dataset is documented.
Industry/context	French cell-phone industry [1]	State clearly; do not describe as an Australian survey.
28-item thesis mapping	Not evidenced in supplied files	Attach complete item-level mapping and rationale.
TDM direction	Unresolved	Verify wording/anchors/coding; document any reverse coding.
Repeated complete profiles	Not auditable without dataset	Report unique rows, repeated full profiles, IDs/independence checks.
Data rights/licence	Not evidenced in supplied files	Record source URL/version/retrieval date and permitted use.

a good evidence for a multidimensional index. Rather, the high correlations and VIFs demand that the study rethink the issue of construct distinctness and that the results of coefficient changes not be taken as independent effects. This finding directly influenced the design that is used in Chapter 6, which is a measurement first design.

Based on the authors' experience, the first constraint is construct provenance. The item-to-construct mapping is not a result of the original construction of the benchmark dataset, thus the mapping needs to be validated instead of presumed. The second is that constructs overlap considerably, limiting interpretation of the independent effects of constructs. The third is that the practical validation is not yet complete, thus usefulness within the context of organizational decision making is yet to be tested.

MR604 will focus on closure of the evidence in a dependency first basis. The first step is the completion of the Provenance (questionnaire/codebook), item mapping, coding direction and repeated-profile audit. The second phase is to rerun and record the measurement and structural diagnostics with confirmed data and software setup. The third is to actually execute the CVI implementation on tested observation level data and conduct comparator, sensitivity, uncertainty and negative case analyses. The fourth is the practical validation (if feasible) of the proposed solution with ethics and access requirements. The last step is thesis integration, cross reference checking, reference auditing and preparation of the publishable output in defense mode. Publication plan is embedded in thesis. Only a contribution that is sufficiently developed methodologically should be used as the basis for a manuscript. A good first paper will be based on the CVI framework, the logic behind the use of measurement first validation, and empirical lessons about construct overlap and the transparent composite scoring. Any claims made in the manuscript should be those that can be substantiated at the manuscript's date of submission.

The study ends at MR603 level, leading to an improved and better substantiated research stance. The CVI offers a framework to link evidence of software architecture, technical debt, API maturity, readiness of the organization and platform governance and digital transformation capability to an interpretable readiness profile, whereas the CPI is not part of the readiness aggregate but is rather an outcome.

6.5 Measurement and Structural Validation Gates

Gate	Predefined rule	Action if failed
G1 Instrument/provenance	All retained items have verified source, wording, anchors, mapping and coding	Do not finalise construct scoring.
G2 Reliability/convergent validity	Report loadings, alpha/rho_A/CR and AVE; justify thresholds using method literature rather than selectively	Revise/withhold weak construct; document decision.
G3 Discriminant validity	Report HTMT with confidence inference where available; inspect construct correlations and cross-loadings as diagnostics	Restrict separate-construct claims; consider theoretically justified merge/re-specification.
G4 Collinearity	Report inner VIF; values materially above conventional guidance trigger instability review [3]	Do not use unstable path coefficients as weights.
G5 Comparator/robustness	Alternative method must improve predefined stability/predictive criteria without sacrificing interpretability	Retain simpler baseline if no defensible improvement.
G6 Practical validation	Independent users/cases rate clarity, relevance and actionability under approved protocol	Do not claim practical superiority until completed.

The CVI is a sequential gated CV emitter. Errors do not affect the validity of the study in any way, except that they limit what can be concluded [60].

6.6 Corrected Preliminary Evidence and Interpretation

The earlier draft reports construct means of SAQ=3.897, APIM=3.894, CPI=3.851, DTC=3.845, PG=3.824, OR=3.805 and TDM=2.558; correlations with CPI of approximately 0.983–0.994 in magnitude; R^2=0.991 for CPI; and inner VIF values of approximately 50.9–153.4. These values are kept here solely as preliminary results reported as they were only used to provide an overview of the underlying SmartPLS export files which were not provided for independent verification. Interpretation is conservative and is deliberately so. The high value of R2 (0.991) is indicative of a good explanatory fit in the sample, but the very high values of the correlations and VIF suggest that there is a great deal of overlapping/collinearity and that individual path effects are not reliable weights for indexes. The next step is to re-establish measurement distinctiveness, and then proceed to structural interpretation that will not be quite as strong.

6.7 CVI Construction and Reproducible Reference Implementation

If the scale is a validated 5-point construct scale then the normalization rule is: Normalized score = ((construct score − 1) / 4) × 100 If and only if TDM item semantics confirm that the higher the raw TDM the more difficult the task/more prepared the student, then the directionally opposite version of the TDM items 1-5 scale is used prior to normalization. CPI is not included in the readiness aggregate since it is the market product. Equal-weight reference CVI for observation i is the average of six readiness scores that are eligible, and directionally aligned to that observation. When an item is not met by a required construct, a full CVI is not generated; rather a partial CVI is better and a clear warning provided. When there is no defensible empirical basis for differential weights, equal weighting is used as a baseline that is transparent in line with the composite-indicator approach [61]. This revision was implemented in Python, and tested using 5-unit tests. Tests include: scale boundaries, a middle of range case, demonstration of "missing-construct" rejection, demonstration of "out-of-range" rejection, reproduction of the previously reported sample-means demonstration. All 5 tests were successful. The demonstration, when applied to the reported means, and the provisional TDM reversal, returns 69.6125/100. This validates the arithmetic – not the

mapping of the construct, the threshold, the prediction or the commercial readiness of the construct.

6.8 Implementation Procedure and Definition of Done

Step	Input	Action	Output / acceptance evidence
1. Freeze research logic	Approved Chs. 1–5	Copy exact RQs/ROs/hypotheses; log changes	Signed/dated change record; no identifier drift.
2. Audit data/instrument	250-row file + questionnaire	Check provenance, row independence, missingness, item map, TDM coding	Audit report + codebook + mapping matrix.
3. Re-run measurement model	Verified data/project	Run final SmartPLS settings; export all diagnostics	Native/exported tables with version/settings.
4. Apply gates	Measurement/structural outputs	Apply G1–G6 decision rules	Gate register with pass/fail/restrict decisions.
5. Compute CVI	Eligible observation-level scores	Run versioned Python implementation	Record-level scores + warnings + checksum/version.
6. Test robustness	Baseline + comparator	Sensitivity, uncertainty, negative cases, holdout/prediction if defensible	Predefined metrics and results table.
7. Practical validation	Approved participants/cases	Run structured rubric after ethics/access approval	Anonymised evidence + summary + limitations.
8. Freeze final artefacts	All outputs	Tag repository and archive settings/logs	Commit,testreport thesis appendices.

6.9 Predefined Evaluation, Sensitivity and Decision Rules

Area	Metric/procedure	Predefined decision rule
Baseline	Equal-weight CVI	Reference model; never called "validated" solely from arithmetic reproducibility.
Comparator	One theoretically defensible alternative weighting method	Adopt only if measurement gates pass and improvement is stable, interpretable and not driven by collinearity.
Sensitivity	Vary admissible weights ±10% around baseline and conduct leave-one-dimension-out stress test	Flag instability if conclusions/rank ordering change materially; report rather than conceal.
Uncertainty	Bootstrap relevant model quantities where methodologically valid	Report confidence intervals and avoid unsupported decimal precision.
Predictive evidence	Holdout/PLSpredict-style assessment where data independence permits [3]	Do not claim predictive usefulness without out-of-sample evidence.
Negative cases	Missing construct; failed validity gate; wrong TDM direction; extreme imbalance	System must block full score or issue explicit warning as specified.
Thresholds	No commercial-readiness cut-off is currently validated	Do not create green/amber/red cut-offs until empirical/practical validation supports them.

6.10 Ethics, Data Governance, Security and Responsible AI

No ethics approval was provided or an expert-validated dataset was not provided, nor was there any completed ethics approval. Therefore, further validation in practice is still to be planned. Prior to the collection of participants, expert or organizational data, the study should include documentation of the ethical pathway that will be followed, the process for recruitment, consent, withdrawal, confidentiality, de-identification, access to the data, storage location, length of time for retention, and data disposal. In case of secondary data, the rights of the source and the allowable use criteria need to be documented. The use of AI or automatic tools should not replace verifying sources or analyzing data, or ownership by the student. The use of AI to draft/code should be reviewed for accuracy with sources and institutional requirements. This revision does not include any fabricated results, participants, approvals or software output.

6.11 Risks, Scope Control and Chapter Synthesis

Risk	Consequence	Control
Unverified ECSI→CVI mapping	Threatens construct validity	Resolve before final inferential claims; restrict conclusions if unresolved.
Repeated-profile/independence issue	Can distort standard errors and apparent relationships	Run row/profile audit and document provenance.
TDM direction uncertainty	Can change CVI score materially	Use item semantics/coding, not correlation sign, to decide direction.
Severe construct overlap/VIF	Undermines distinct dimensions and path weights	Reassess measurement model; avoid path-derived weights.
Dashboard scope creep	Consumes validation time	Defer until core evidence and thesis are complete.
Publication pressure	Encourages premature claims	Keep manuscript secondary to thesis validation.

Chapter 6 thus establishes a tested and repeatable pathway without being overly prescriptive, while adhering to current evidence. The best artefact that is fully implemented in this revision is the transparent implementation of reference scoring and the tests. The biggest risk that remains to be answered is not coding – it is the validity of the constructs of the thesis being represented by the benchmark indicators. This question needs to be answered first before the CVI can be considered an empirically validated.

Chapter 7: Conclusion, Limitations, Future Work and Publication Plan

7.1 Research Summary and Progress to Date

This research attempts to bridge the technical and commercial gaps between working prototypes of digital assets and the overall conditions for the commercial development of digital-asset platforms. To date, the work has resulted in a candidate six-dimension CVI architecture, Measurement First Validation Gate Logic, Capturing the underlying Equal Weight baseline, Clear Risk controls, and a Reproducible reference implementation. The existing evidence also shows that there is a big methodological problem, namely that construct correlations and VIF values are very high, so the apparent good fit of the structure to the data is not enough to justify assuming distinct readiness dimensions and to justify stable path-derived weights [62].

7.2 Current Contribution and RQ/Objective Status

Research element	**Current status**	**Defensible contribution now**	**Evidence still required**
Factors/readiness dimensions	Proposed / partially evidenced	Coherent candidate structure from prior thesis work	Verified item provenance and construct mapping.
Relationships among constructs/CPI	Preliminary	Shows why collinearity/validity must be treated explicitly	Final verified SmartPLS outputs and measurement decisions.
CVI construction	Implemented as reference calculation	Transparent normalisation, gating and equal-weight logic	Verified observation-level dataset and final TDM direction.
Evaluation/applicability	Protocol defined	Predefined robustness and failure rules	Independent predictive/practical validation.
Decision support	Designed	Dimension profile + evidence warnings	User/case evaluation after ethics/access approval.

7.3 Limitations and Consequences for Interpretation

Limits are functional, they're not pretty. First, the benchmark is a dataset from an ECSI mobile-phone, rather than a dataset originally created for measuring the digital-asset readiness constructs of the thesis. The major threats are threat to construct validity without an item level mapping. Second, there is no evidence of the second 28 item questionnaire/code book and TDM coding. Third, correlation values and VIFs reported were very high, suggesting possible construct overlap and unstable structural coefficients. Fourth, repeated-profile and

row-independence claims can't be audited without the underlying data. Fifth, there was no ethics approval evidence provided and it was not validated in practice. These restrictions do not reflect a validated commercial-readiness threshold, a firm score or a causal result or predictive claim.

7.4 Executable MR604 Completion Plan

Phase	Deliverable	Dependency	Completion criterion
1. Evidence freeze	Authoritative RQ/RO list + provenance pack	Supervisor-approved prior chapters	No unresolved wording/sample-size contradictions.
2. Instrument/data audit	Questionnaire, codebook, 28-item map, duplicate/profile audit	250-row dataset and source records	Every item/row traceable; TDM direction resolved.
3. Measurement revalidation	Final reliability/validity/collinearity report	Audited instrument/data	Explicit retain/revise/merge/drop decisions.
4. Reproducible implementation	Versioned Python package + environment + tests	Eligible constructs	All acceptance tests pass; record-level output reproducible.
5. Robustness/prediction	Comparator, sensitivity, uncertainty, negative-case results	Stable measurement model	Protocol completed and deviations logged.
6. Practical validation	Approved expert/case evaluation	Ethics/access	Evidence collected/anonymised and analysed per protocol.
7. Thesis integration	Final Chapters 1–7 + appendices + reference audit	All core evidence complete	100% citation/reference match; all claims linked to artefacts.
8. Optional dashboard/publication	Prototype/manuscript only if time remains	Core thesis complete	Does not delay validation or submission.

7.5 Publication Plan

Publication is regarded as an output of an advanced thesis contribution without being considered as validation. The first manuscript should address the measurement-first CVI framework and the implications of construct overlap/collinearity for index weighting, as well as the findings of the empirical studies that have been conducted with this validated approach to the measurement once the problems of provenance and measurement are resolved. Only after scope, type of article, word limits and current guidelines for authoring are matched, a target outlet should be selected. Authorship and contribution statements should be done prior to submission, in consultation with the supervisor. None of the conference plans are accepted in this report, they are only “planned” until a status of official call and submission are confirmed.

7.6 Final Conclusion

With the revised thesis direction, it is a narrower thesis that is more defensible. The CVI is offered as a research artefact that can be tested and is built with explicit evidence gates and not as an instrument ready for market. The first step is to establish the "provenance" and "construct validity" and then reproduce the analysis and scoring from records that can be audited, and then, only assess the robustness and practical use. This sequence directly tackles the issues of evidence, integrity, reproducibility and scope that the examiner is concerned with. MR604 can transform the current framework from a believable design to a contribution that is validated and in which the pre-defined gates are met without having to use unsupported claims.

Chapter 8: Plans for Future Publication

8.1 Chapter Introduction

In the following chapter, a realistic course of events is outlined for publishing the thesis research "From Prototype to Platform: Measuring the Commercial Viability of Digital Assets" as peer-reviewed research outputs. The publication strategy is deliberately connected to evidence developed in MR603, and to the implementation and validation work to be undertaken in MR604. Does not take preliminary results as final. Rather, publication will be delayed until the data provenance, measurement model, implementation of the CVI, and

robustness checks and evaluation in practice are sufficiently developed. The main publication channel of the supervisor is the International Conference on Emerging Technologies and Education 2027 (ICETE 2027). According to the official conference information, the full papers are about 6-10 pages long, the paper submission deadline is 1 November 2026, the notification is on 1 December 2026, the camera-ready deadline is 4 January 2027, and the conference dates are 8–9 February 2027. These details will be again checked just before submission as the requirements of the conference may change.

8.2 Publication Objectives

The publications activity has three aims. First, it will present the main research contribution, a measurement-based Commercial Viability Index (CVI) framework to determine if software based digital assets are ready to be taken from prototype to scalable platform. Second, it will be used to record the implementation evidence and evaluation evidence created in MR604, such as Python-based scoring, robustness analysis and the planned decision-support dashboard. Third, it will give clear scholarly documentation of the study and will keep track of the connections among the research questions, research methods, results, limitations, and final claims.

Publication objective	Connection to thesis	Evidence required before submission
P1. Present the CVI framework	RO1–RO4; RQ1–RQ3; Chapters 1–6	Stable construct definitions, verified coding, defensible CVI architecture and scoring logic
P2. Demonstrate implementation and evaluation	RO5–RO6; RQ4; MR604 implementation	Reproducible Python implementation, final SmartPLS evidence, comparator/sensitivity results and practical evaluation where approved
P3. Translate the contribution into publishable research	Chapter 8 and final thesis synthesis	Clear manuscript contribution, ethical data use, supervisor review, verified venue requirements and complete references

Table 8.1. Publication objectives and evidence requirements

8.3 Primary Publication Venue: ICETE 2027

The published themes of ICETE 2027 are Data Analytics, Business Analytics and Digital Transformation, and Management, Innovation and Entrepreneurship making it the first choice of the attendees. These areas offer a straight path to place the CVI as a framework for data-informed decision support for digital transformation and commercialization. The paper will not be a general software metric, but will be more of an integrated assessment of

Business Analytics and Digital Transformation	The study evaluates readiness for moving digital assets from prototype toward scalable platforms.	Digital transformation capability, organisational readiness and evidence-based commercialisation decisions.
Management, Innovation and Entrepreneurship	The framework supports technology managers, startups and decision makers evaluating platform investment readiness.	Practical interpretation, risk warnings, innovation readiness and commercialisation implications.

technical and organizational readiness, and the evidence needed before a digital asset can be considered a commercial product for platform evolution.

Table 8.2. Alignment of the thesis with ICETE 2027 themes

8.4 Proposed Paper 1: ICETE 2027

Proposed paper title: From Prototype to Platform: A Measurement-First Commercial Viability Index for Digital Assets

Primary target: International Conference on Emerging Technologies and Education 2027 (ICETE 2027)

Paper 1 will introduce the core thesis contribution: a measurement first Commercial Viability Index (CVI) which will evaluate if a software-based digital asset is technically and organizationally ready to move from prototype to a viable scalable platform. The manuscript will highlight the six readiness dimensions, the CVI architecture with the validation gated, the 0–100 scoring logic which is transparent and the methodological choice not to transform unstable PLS-SEM path coefficients into index weights when there is severe multicollinearity. It will only cite evidence verified at the time of the submission and will clearly distinguish between preliminary MR603 evidence and MR604 validation work.

Keywords: This study uses the Commercial Viability Index (CVI), digital assets, platform readiness, PLS-SEM, and digital transformation.

Alternative publication titles

1. Design and Validation of a Commercial Viability Index for Digital Asset Readiness for Platform Evolution.

2. Beyond Technical Feasibility: A Validation-Gated Commercial Viability Framework for Digital Assets by Anna Gadoff, CTI's VP of Product.Beyond Technical Feasibility: A Validation-Gated Commercial Viability Framework for Digital Assets by Anna Gadoff, VP of Product at CTI.

3. A Measurement First Framework for Assessing Prototype to Platform Readiness of Digital Assets

8.5 Proposed Paper 2: ACM TRUST 2027

Proposed paper title: Trustworthy Commercial-Readiness Assessment for Digital Assets: Reproducible CVI Scoring, Assurance and Decision Support

Primary Target: ACM Conference on Trustworthy and Responsible AI and Computing Systems (ACM TRUST 2027)

Paper 2 will not be a repeat of the implementation-and-assurance study in Paper 1. Its contribution will be centered on trustworthy decision support, which means reproducible Python-based CVI scoring, explicit evidence and validation gates, uncertainty and sensitivity analysis, data/provenance controls, auditable warnings and the planned prototype dashboard. This framing captures the most closely the ACM TRUST themes related to test and evaluation, benchmarking and assurance, transparency and auditability, data governance, and responsible computing. The paper will only be attempted if sufficient original implementation / evaluation evidence will be provided by MR604, otherwise it will be a future paper publication target and not a submitted claim.

Keywords: Trustworthy Computing, decision support, Commercial Viability Index, reproducibility, assurance, data governance.

Alternative publication titles

1. commercial viability assessment framework for digital assets, which can be reproduced using python tools.
2. Evidence-Gated Decision Support for Digital Asset Readiness – Robustness, Reproducibility and Assurance.
3. 3. Trustworthy Prototype to Platform Decisions: A Commercial Viability Index based on Assurance.

8.6 Publication Ethics and Distinction Between the Two Papers

The two-paper approach will be employed only when the final research paper has sufficient original content to provide two legitimate scholarly contributions. Paper 1, and Paper 2, will

not be perceived as two versions of the same manuscript. They will focus on different aspects of the research, be different evidence packages, and they will be principal contributions – and no two manuscripts will be submitted at the same time to more than one place. Essential background and methods which are shared will be disclosed and cited where appropriate and rewritten if necessary.

Publication dimension	**Paper 1 – ICETE 2027**
Central question	How can technical and organisational readiness be integrated into a defensible Commercial Viability Index for prototype-to-platform decisions?
Primary contribution	Measurement-first CVI framework, construct logic, validation gates and transparent baseline scoring.
Main evidence	Verified measurement model, PLS-SEM diagnostics, CVI architecture and robustness evidence available at submission.
Audience	Digital transformation, analytics, innovation and technology-management researchers.
Distinctive emphasis	Framework design and empirical measurement of commercial/platform readiness.

Table 8.3. Distinct publication identity of Paper 1

Publication dimension	Paper 2 – ACM TRUST 2027
Central question	How can CVI-based digital-asset readiness decisions be implemented so that scoring, warnings and evidence are reproducible, auditable and appropriately assured?
Primary contribution	Python implementation, assurance logic, robustness/uncertainty checks, provenance controls and decision-support prototype.
Main evidence	Version-controlled code, worked cases, test outputs, sensitivity/uncertainty analysis, audit trail and practical evaluation where ethically permitted.
Audience	Trustworthy/responsible computing, assurance, evaluation, governance and systems researchers.
Distinctive emphasis	Implementation trustworthiness, reproducibility, auditability and assurance rather than re-presenting the framework paper.

Table 8.4. Distinct publication identity of Paper 2

Publication-integrity controls

Authorship will be based on the actual contribution by the authors and the agreement of the institutions. There will be no fabricated, duplicated or selectively reported evidence used. It will only include participant, expert or organization evidence where ethics, consent, data rights and privacy and data retention requirements are met. Data on which the restriction has been placed will not be made public on data repository. Where manuscripts overlap, this will be indicated, and where possible, work will be cross-cited and there will be no excessive repetition of text, tables, figures or results. All conference requirements, deadlines and templates will be rechecked just before it is submitted.

8.7 Planned Structure of the Priority Conference Paper

The conference manuscript will not be copied from the thesis, but will be derived from it. The methodology, CVI architecture, implementation, results and evaluation will be the strongest spaces, and will show the original contribution. Background material will be summarized to a literature of problem and gap.

Paper section	Planned content	Evidence source
Title, Abstract and Keywords	Problem, objective, method, principal results and contribution in concise form.	Final evidence freeze before submission
1. Introduction	Prototype-to-platform problem, motivation, gap, research objective and contribution.	Chapters 1–2
2. Related Work	Software architecture, technical debt, API maturity, organisational readiness, governance, digital transformation and existing assessment approaches.	Chapter 2
3. Methodology	Research design, data provenance, construct specification, PLS-SEM settings, validation gates and CVI scoring rules.	Chapters 3–6

4. CVI Framework and Implementation	Five-layer architecture, 0–100 normalisation, equal-weight baseline, reproducibility workflow and Python implementation.	Chapter 6 + MR604 implementation
5. Results and Evaluation	Measurement/structural results, collinearity, comparator, sensitivity, uncertainty and practical evaluation where available.	Chapters 4–6 + MR604
6. Discussion	Meaning of results, comparison with prior research, practical implications and answer to the publication research question.	Final thesis discussion
7. Limitations and Future Work	Provenance, generalisability, measurement limitations, dashboard/case validation and future research.	Chapters 7–8
8. Conclusion	Concise statement of validated contribution without introducing new evidence.	Final thesis conclusion

Table 8.5. Planned structure of the priority conference paper

8.8 MR604 Work Supporting Publication

MR604 will fill the gap between a framework paper and a validated implementation of the publication. Therefore, the publication tasks will proceed simultaneously with the implementation of the theses, not instead it. Once the CVI core model is fully stable, the three MR604 goals are to complete and test the empirical CVI model; implement and test the CVI scoring framework; and create a Python-based CVI dashboard and prototype of a decision-support system.

MR604 phase	Research / implementation task	Publication output
Weeks 1–2	Resolve provenance, questionnaire/codebook, coding, repeated-profile audit and TDM direction.	Verified methods and data-provenance notes.
Weeks 3–4	Finalise measurement specification and clean/version the analysis dataset.	Stable construct and measurement description.
Weeks 5–6	Run final SmartPLS measurement and structural analysis.	Final tables for reliability, validity, VIF, bootstrapping and structural results.
Week 7	Implement observation-level equal-weight CVI in Python and maintain version-controlled GitHub files.	Reproducible formula/script, worked cases and tests.
Week 8	Run comparator, sensitivity and uncertainty analysis.	Robustness/evaluation section and figures.
Weeks 9–10	Conduct practical/case evaluation where ethics and access permit.	Practical evidence and interpretation.
Weeks 9–11	Develop the Python/Streamlit dashboard after the core model is stable.	Decision-support prototype and visual outputs.
Weeks 11–12	Integrate results, discussion, limitations and paper draft.	Near-final manuscript for supervisor review.
Weeks 13–14	Supervisor review, reference audit, formatting and submission checks.	Submission-ready thesis and publication package.

Table 8.7. MR604 research-to-publication work plan

8.9 Submission Timeline and Decision Points

According to the current official call, paper submission for ICETE 2027 will be 1 November 2026 which is provided by the supervisor in the plan. The decision to submit will be determined by the maturity of the evidence that is available, as this is before the full MR604 implementation cycle. Only a paper which can make a coherent and defensible contribution

Date / stage	Planned action	Decision criterion
September 2026	Confirm paper scope, ICETE themes, title and manuscript outline.	Supervisor agreement on contribution and venue fit.
Early October 2026	Complete first integrated draft using verified MR603 evidence.	No unsupported claims; clear methods and limitations.
Mid–late October 2026	Supervisor review, technical revision, reference and formatting audit.	Evidence and paper quality sufficient for external submission.
1 November 2026	Target ICETE 2027 submission, subject to readiness.	Submit only a complete, defensible manuscript.
1 December 2026	Expected ICETE notification according to current CFP.	If revision is requested, address reviewer comments systematically.
4 January 2027	Current camera-ready date for accepted ICETE papers.	Final compliance, authorship, formatting and evidence checks.
8–9 February 2027	Current ICETE 2027 conference dates.	Present accepted work; publication acceptance is not required for thesis completion.
MR604	Continue implementation/evaluation regardless of conference outcome.	Thesis quality and validation remain the primary research priority.

at that time will be submitted. Any additional evidence produced in MR604 will not be presented as being already available, but instead will be presented in the thesis (extended paper) or in a second publication.

Table 8.8. Proposed publication timeline and decision points

8.10 Second Publication and Contingency Strategy

The second publication should occur once the contribution in the Python implementation, robustness analysis and dashboard/case evaluation has been made that is truly different from Paper 1. This second paper would likely be aimed at ACM TRUST 2027, which will have trustworthy and responsible computing, governance, evaluation, benchmarking and assurance as part of the technical program. All submissions to the final track, manuscript format, final submission deadline and eligibility will be double checked with the official call. Paper 2 will be submitted for other appropriate peer-reviewed venue after being compared by the supervisor, not the thesis-critical validation work will weaken. If a thesis is rejected or if there are timing issues, it will not be delayed in completing the process. The first manuscript will be rejected if it does not meet the criteria for publication, and suggestions from the reviewers will be incorporated in the manuscript to improve it and it will be sent to appropriate journals. The thesis will be the main product and any optional scope for publication or dashboard will be cut down before the essential validation, reproducibility, writing or supervision review process is jeopardized.

8.11 Authorship, Ethics, Data and Research Integrity

This thesis will lead to manuscripts that will be authored from the student side by Mahmud Jamil. Co-author inclusion will be based on true scholarly contributions, institutional requirements/agreements prior to submission. Where necessary, ethics approval, consent, data rights and privacy conditions will be met and so will participant or expert or organizational data be included. Restricted data will not be made public or placed in a public repository in GitHub, but appropriate code, data dictionaries and data reproducibility instructions will be made public where sharing is not permitted. The two-paper approach will be based on the principles of publication-integrity. A single paper will not be sent out to several journals at once; the two papers will not duplicate anything but a few words of text or results; related manuscripts will be disclosed and cited when necessary. References will be

verified for accuracy with the source(s) and will be cited consistently in the appropriate IEEE format. In all manuscripts, all statistical, implementation and visual results reported will be traceable to the analysis outputs, analysis scripts or documented analysis research records retained.

8.12 Thesis Contribution-to-Publication Mapping

The two proposed publications are directly linked to the thesis research questions, objectives, and implementation outputs and evidence requirements to strengthen the traceability. This mapping will minimize the chances of treating publication as a separate activity from the thesis and be clear about what can be demonstrated at each phase of the research.

Thesis contribution / evidence	Paper 1 – ICETE 2027	Paper 2 – ACM TRUST 2027
Integrated six-dimension CVI architecture	Primary contribution: framework design and prototype-to-platform readiness logic.	Background foundation only; not repeated as the principal contribution.
Measurement model and validation gates	Primary empirical evidence: reliability, validity, collinearity and defensible scoring decisions.	Used as assurance inputs that determine whether system outputs can be trusted.
Equal-weight baseline and comparator logic	Explains why transparent baseline scoring is retained when structural weights are unstable.	Supports reproducible and auditable decision-support behaviour.
Python CVI implementation and GitHub record	Supporting reproducibility evidence.	Primary implementation contribution: traceable scoring, testing and auditability.
Sensitivity, uncertainty and negative-case checks	Evidence of robustness of the proposed index.	Primary assurance evidence for trustworthy system behaviour and warning generation.
Dashboard and practical evaluation	Secondary demonstration where space and evidence permit.	Primary system-facing evidence: transparent outputs, warnings, human interpretation and practical evaluation.

Table 8.9. Thesis contribution-to-publication mapping

8.13 Research-to-Publication Readiness Framework

The publication process will be based on the evidence gating. Only when the research stage before the progression to an external manuscript will be well documented will progression to an external manuscript take place. This helps to ensure that this study does not begin to be claimed as finished and that publication activity does not become more important than the quality of the thesis evidence.

1. Evidence foundation	2. Validation	3. CVI implementation	4. Paper 1	5. Assurance extension	6. Paper 2
Verified provenance, coding and construct definitions	Measurement, structural and robustness checks	Python scoring, tests and reproducibility record	CVI framework and empirical validation	Sensitivity, auditability, dashboard and practical evaluation	Trustworthy implementation and assurance contribution

Figure 8.1. Evidence-gated research-to-publication pathway

In the case of Paper 1, the minimum readiness gate is a dataset/instrument that has been verified, a defensible construct specification, final measurement diagnostics, and CVI scoring that is transparent, with adequate evidence of robustness to support claims that are presented in the manuscript. The minimum gate for Paper 2 is stronger; for its Python implementation, the program must be stable, testable and traceable, have documented sensitivity or uncertainty behavior, have warning logic, have a record of reproducing and have a practical example.

8.14 Expected Publication Contribution

It is not merely a new numerical index that is so expected that scholars are expected to make. Paper 1 aims to bring in a measurement first, validation gated approach for determining when a commercial readiness score is defensible that includes technical and organizational readiness, but maintains CPI as an external output. Paper 2 is designed to build upon that contribution and make it a reliable way to do computing responsibly by making its outputs reproducible with Python, and the warning logic testable, providing sensitivity and

uncertainty evidence, making it auditable, and making its dashboard outputs transparent and interpretable by human readers. The papers are intended to provide a cohesive publication trajectory yet to also be sufficiently unique in the questions being asked, evidence and contribution.

8.15 Chapter Summary

This chapter sets out an evidence-based publication pathway, which is directly linked to the thesis, and the work to be completed for MR604. In paper 1, the focus will be on ICETE 2027 and the measurement-first CVI framework and empirical validation. Paper 2 is aimed at ACM TRUST 2027, and is focused on the implementation of paper 1 with trustworthy outcomes, as well as reproducibility, assurance, auditability and decision-support behavior. The contribution mapping, readiness framework and risk plan are clear on when the evidence to support the decision to publish is mature. The validity of CVI will not be assumed when submitting to or being accepted for a conference and publication activity will not take precedence of the thesis-critical priorities of provenance, validation, reproducibility, practical evaluation, writing and supervisor review.

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